

Historic Rate Rollover FX Forwards

What the technique actually does, why it is effective, what it converts your risk into, and which private credit strategies can carry it.

WHAT IT CHANGES	WHAT IT CREATES	VERDICT
Timing Not the FX risk, not the FX cost. Only when the cash moves — for a fee of 30–50bp.	53% Of hedged notional: the unsecured dealer credit line a 5-year hedge needs at the 99th percentile.	2 of 6 Infrastructure debt and direct lending, conditionally. Not the other four strategies.

THE ONE-SENTENCE VERSION

A historic rate rollover does not reduce FX risk and does not reduce FX cost. It takes the cash settlement of a hedge loss and replaces it with an **uncollateralised loan from your FX dealer**, priced inside an off-market forward rate.

Everything that follows — every benefit, every risk, every suitability judgement — is a consequence of that single substitution.

Prepared for a second-line review of a proposed Historic Rollover Rate FX forward convention.
All figures are computed from an explicit model; assumptions and source code are set out in Appendix B.
Worked example throughout: a EUR-denominated fund holding a USD 100m private credit asset.

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Read sections 1–6 for the mechanics, 11–18 for the risk position, 19–22 for limits and surveillance, 23–26 for the strategy-by-strategy verdict. Section 29 is the question list to take into the approval meeting.

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What the instrument actually is

Six sections that build the trade from nothing. If you already know what an FX forward is, you can start at section 3 — but section 2 contains the one fact most people get wrong about hedging cost.

1 The problem it is built to solve

Start with the problem, not the instrument. A private credit fund that hedges currency risk has a structural cash-flow asymmetry, and it is worth being precise about where it comes from.

Take a EUR-denominated fund that lends USD 100m to a US borrower on a five-year bullet term loan. The fund's investors subscribe in euros and are measured in euros. The asset pays in dollars. Without a hedge, the fund's investors own a five-year dollar bet they did not ask for, roughly ten times the size of the credit spread they are being paid to take. So the fund hedges: it sells USD 100m forward against euros.

An FX forward market is deep out to a year and thin beyond it. Five-year forwards exist, but the bid-offer is wide, the credit charge is material, and — decisively — the fund cannot know that its five-year loan will still be outstanding in five years. Private credit loans prepay, get amended, get extended, get restructured. So in practice nobody hedges a five-year loan with a five-year forward. Everybody hedges it with a three- or six-month forward and rolls it, sixteen or twenty times over the life of the asset.

Now the asymmetry appears. At each roll date the maturing forward settles in cash. If the dollar has strengthened, the fund's short-dollar hedge has lost money and the fund pays that loss in cash, within two business days. Meanwhile the loan — which has gained exactly the offsetting amount in euro terms — pays nothing. It is a five-year bullet. It will not produce a euro of principal until year five, and it cannot be sold, because there is no market to sell it into at anything like carrying value.

THE ASYMMETRY, STATED EXACTLY

A rolling FX hedge on an illiquid asset produces **cash losses against unrealised gains**. The hedge settles on a two-day cycle. The asset settles once, years later. The two are economically offsetting and operationally unrelated.

The magnitudes are not small. A EUR-based fund that is fully hedged into dollars loses roughly 9% of the hedged notional in cash if EURUSD falls from 1.10 to 1.00. That is not a stress scenario; that is 2014–15 (1.39 to 1.05, a 24% move in twelve months) and it is 2021–22 (1.23 to 0.96, 22% in ten months). A fund carrying EUR 300m of hedged dollar exposure would have needed to find something in the order of EUR 60–70m of cash across either of those episodes, out of an asset base that produces cash only on scheduled amortisation and repayment.

Its options are all bad. It can hold a large permanent cash drag, which destroys the return the investors came for. It can draw the subscription line, which is expensive, capacity-constrained and disclosed. It can call capital, which is slow and, if repeated, damages the relationship with the LP base. It can sell assets, which it cannot do at price. Or it can under-hedge, which reintroduces the currency risk it was hired to remove.

The historic rate rollover is a sixth option: **do not pay the cash**.

2 First principles: what an FX forward is

An FX forward is not a bet and not an option. It is a pair of borrowings. When the fund sells USD 100m one year forward against euros, the dealer replicates it by borrowing dollars for a year, selling them spot for euros, and lending the euros for a year. The forward rate is therefore fully determined by the two interest rates — there is no view in it at all.

COVERED INTEREST PARITY

$$F = S \times (1 + r_{\text{USD}} \tau) / (1 + r_{\text{EUR}} \tau)$$

With EURUSD spot $S = 1.1000$, one-year USD at 4.5%, one-year EUR at 2.5%:

$$F = 1.1000 \times 1.045 / 1.025 = 1.12146$$

Read what that means for the fund. It has agreed to buy euros in a year at 1.12146 when it could buy them today at 1.1000. It is paying more for its euros than spot — about 1.95% more. That difference is not a fee and not a spread. It is the interest rate differential, and the fund pays it because it is hedging the higher-yielding currency back into the lower-yielding one.

THE FACT MOST PEOPLE GET WRONG

Hedging a USD loan book back to EUR **costs approximately the USD–EUR interest rate differential, every year, forever**. At today's differential that is 150–200bp. On a direct lending strategy grossing SOFR + 550, the euro investor is handing back a quarter to a third of the spread before a single default. This is a permanent, structural cost of the hedge, and it has nothing whatsoever to do with historic rate rollovers — you pay it either way. It matters here only because it is the reason people go looking for ways to make FX hedging "cheaper", and then reach for a technique that does not, in fact, make it cheaper.

The mechanics of settlement matter for what follows. At maturity, if the fund has sold USD 100m at K , it delivers USD 100m and receives EUR 100m/ K . Its euro profit or loss, marked at the prevailing spot S , is:

SETTLEMENT VALUE OF THE HEDGE, IN EUROS

$$P\&L_{\text{EUR}} = N/K - N/S$$

Negative when $S < K$. The euro buys fewer dollars than the contract assumed, so the contract to buy euros is out of the money. In plain terms: **a strengthening dollar costs the hedge money**.

In almost every case, the fund does not actually deliver USD 100m, because it does not have USD 100m — its dollars are locked inside a loan. It closes out against the market and pays or receives the euro difference. That euro difference is the cash flow the whole technique exists to suppress.

3 What a standard rolling hedge does to your cash

Before looking at the rollover, be clear about the baseline, because the baseline is the honest comparator and it is usually skipped.

Under a conventional rolling hedge, at every roll date the fund does two independent things: it settles the maturing forward in cash, and it strikes a new forward at the prevailing market rate. There is no memory. Each new contract starts at fair value zero. The strike on the fund's books always looks like a market rate, because it is one.

Run the five-year loan through a persistent dollar rally — EURUSD from 1.10 down to 0.93, which is a 15% move spread over five years and well inside historical experience:

TABLE 1 · STANDARD ROLLING HEDGE, SCENARIO A

ROLL	EURUSD SPOT	STRIKE SETTLING	CASH SETTLED (EUR M)	CUMULATIVE CASH (EUR M)	NEW STRIKE
Year 1	1.0500	1.1215	−6.07	−6.07	1.0705
Year 2	1.0000	1.0705	−6.58	−12.65	1.0195
Year 3	0.9700	1.0195	−5.01	−17.66	0.9889
Year 4	0.9500	0.9889	−4.14	−21.80	0.9685
Year 5	0.9300	0.9685	−4.28	−26.08	—

USD 100m hedged. Negative figures are cash out of the fund. The strike is re-struck at the market forward at every roll, so it tracks spot down.

Two observations. First, the fund pays out EUR 26m over five years — a quarter of its euro capital — and this is the hedge *working correctly*. The loan has gained an almost exactly offsetting amount in euro terms. Second, the fund had to find EUR 4–7m of cash on five separate occasions, with nothing coming back from the asset until year five. That is the problem. Note also that the strike on the books is always a plausible market rate; nothing here would ever trip an exception report.

4 The historic rate rollover, step by step

Now the substitution. At the roll date, instead of settling the maturing forward against the market, the fund books a two-legged FX swap in which the near leg is a mirror image of the maturing contract — same notional, same value date, and critically **the same strike**.

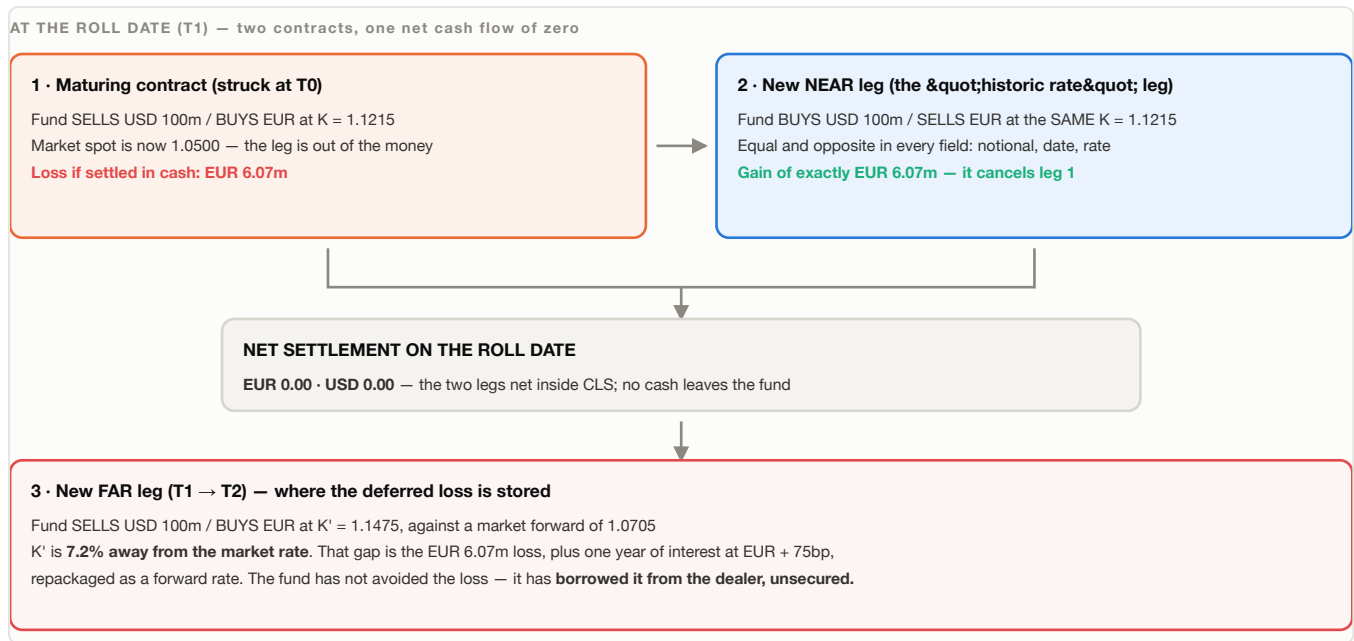
The maturing contract and the new near leg net to nothing. Same currencies, same amounts, same day, opposite directions. They settle against each other inside CLS and no cash moves in either direction. That is the entire trick, and it is genuinely elegant: the loss has not been paid because the two obligations that would have produced the payment have been made to cancel.

But a dealer does not do this for nothing, because the near leg is off-market. At the roll date, buying dollars at 1.1215 when spot is 1.0500 is a gift worth EUR 6.07m — and the dealer has just given it. The dealer recovers it on the far leg, which is deliberately struck away from the market by exactly the amount of the deferred loss plus interest for the roll period. The far leg is where the loss goes to live.

FIGURE 2

The exact mechanism: two off-setting legs, zero net cash, one hidden loan

Scenario A, first roll — EURUSD has fallen from 1.1000 to 1.0500



THE RECURSION, IN FULL

Let D_t be the deferred balance (positive = fund owes the dealer), c the accrual rate the dealer charges, τ the roll period, F the market forward and K the booked strike.

Balance rolls forward:

$$D_t = D_{t-1}(1 + c \tau) + [N/S_t - N/F_{t-1}] \quad \leftarrow \text{prior balance with interest, plus this period's FX move}$$

New strike is solved from the balance:

$$N/F_t - N/K_t = D_t(1 + c \tau) \Rightarrow K_t = N \div [N/F_t - D_t(1 + c \tau)]$$

Everything in this document follows from those two lines. The second one also contains a hard structural limit that we return to in section 13: as D_t approaches N/F_t , the denominator goes to zero and K_t goes to infinity. Beyond that point the deferred loss cannot be expressed as a forward rate at all and the structure simply stops working.

Applying the recursion across the same five-year dollar rally gives the picture the fund's treasurer will see:

TABLE 2 · HISTORIC RATE ROLLOVER, SCENARIO A

ROLL	SPOT	STRIKE SETTLING	CASH SETTLED	DEFERRED BALANCE (EUR M)	NEW STRIKE BOOKED	MARKET FORWARD	STRIKE VS MARKET
Year 1	1.0500	1.1215	0.00	6.07	1.1475	1.0705	+7.2%
Year 2	1.0000	1.1475	0.00	12.85	1.1790	1.0195	+15.6%
Year 3	0.9700	1.1790	0.00	18.28	1.2158	0.9889	+22.9%
Year 4	0.9500	1.2158	0.00	23.01	1.2580	0.9685	+29.9%
Year 5	0.9300	1.2580	-28.04	—	—	0.9481	—

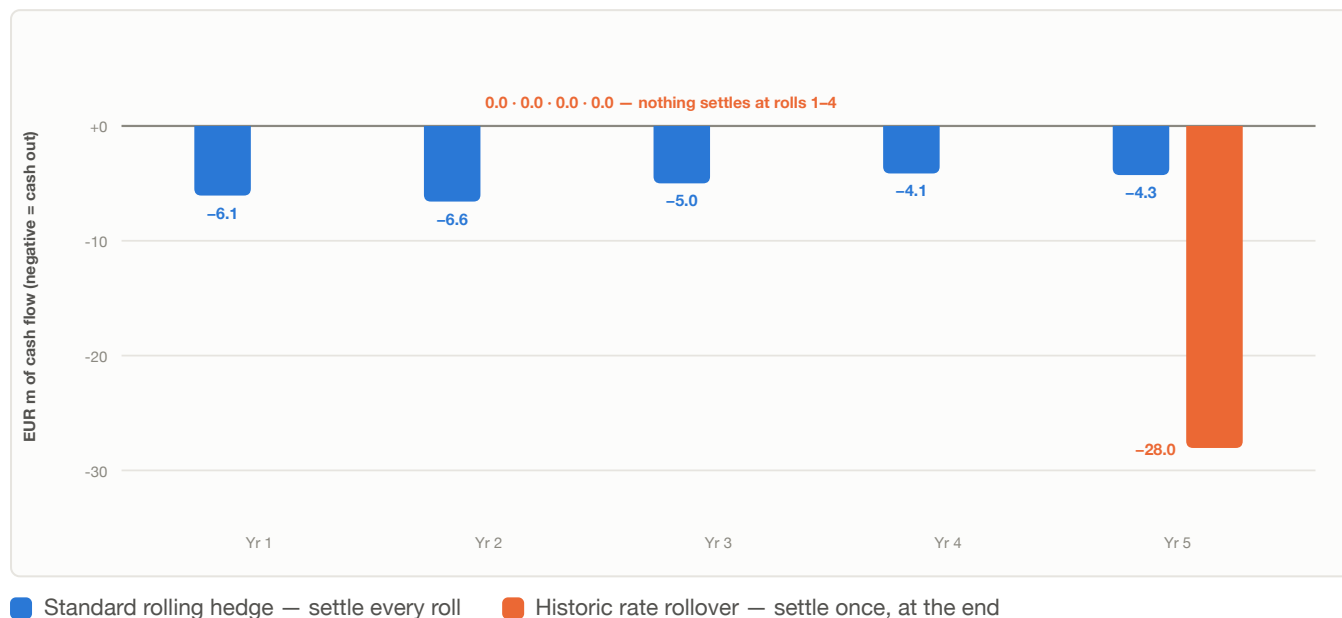
Deferred balance accrues at EUR 2.5% plus a 75bp dealer credit spread. In year five the loan repays and the whole balance settles at once.

Four years of zeros, and then a single EUR 28m payment. That is the trade. Whether it is a good one depends entirely on whether that year-five payment lands on the same day as a year-five repayment of comparable size — and on whether the dealer is still willing to be there in year four.

FIGURE 3

Same economics, opposite cash-flow shape

Scenario A · USD 100m hedged · EURUSD 1.10 → 0.93 over five years



5 Reading the trade blotter: the worked example decoded

The deck's own worked example is worth walking through carefully, because it uses round numbers that obscure the mechanism and it contains one presentational inconsistency that a reviewer should catch.

TABLE 3 · THE DECK'S EXAMPLE, AS BOOKED

	LEG	TRADE DATE	MATURITY	EUR M	USD M	ALL-IN RATE	SPOT REF	MTM \$
Inception	Near	01/01/2026	08/01/2026	10	-11	1.1	1.1	0
	Far	01/01/2026	08/01/2027	-10	12	1.2	1.1	0
Rollover	Near	01/01/2027	08/01/2027	10	-12	1.2	1.1	1
	Far	01/01/2027	08/01/2028	-10	13	1.3	1.1	-1

Reproduced from the source material. Note the narrative line: "near leg equal and opposite of the maturing FWD trade; FWD rate of the far leg reflects unrealized MTM; New Spot Ref is 1.3."

Trace the identity that makes the whole thing work. The far leg struck at inception is *sell EUR 10m / buy USD 12m at 1.2, maturing 08/01/2027*. The near leg of the rollover is *buy EUR 10m / sell USD 12m at 1.2, maturing 08/01/2027*. Every field matches and every sign is reversed. On 8 January 2027 the two settle against each other for exactly nothing: EUR +10 -10 = 0, USD -12 +12 = 0.

Spot has moved to 1.3, so the maturing far leg — selling euros at 1.2 into a 1.3 market — is worth USD 1m less than the market. The near leg is worth USD 1m more. Hence the +1 and -1 in the MtM column. They cancel, no cash moves, and the new far leg at 1.3 carries the position forward.

ONE THING TO FIX BEFORE THIS GOES TO COMMITTEE

The "Spot reference" column shows 1.1 on all four rows, while the narrative states the new spot reference is 1.3. Both cannot be right, and the reader cannot verify the arithmetic without knowing which. More substantively, the example uses a one-week near leg and a one-year far leg with rates of 1.1, 1.2 and 1.3 — round numbers that make the mechanism legible but make it impossible to see the two things a risk reviewer needs to see: **the accrued financing charge** embedded in the far-leg rate, and **the cumulative deferred balance** after the second, third and fourth rolls. A one-roll example always looks benign. The technique's risk profile only becomes visible over several rolls, which is why every table in this document runs five.

The second thing worth extracting from the deck is what it says about governance, because it is more revealing than the mechanics. The stated process is: the portfolio manager decides whether a rollover is necessary; the portfolio manager obtains operations approval; the portfolio manager sends the order with a specific reason code; the dealing desk executes; the portfolio manager checks the executed details are as expected. The line "Historic rollovers are not automatic; the Investment team will determine whether a rollover is necessary at each rollover date" is the load-bearing sentence in the entire pack.

THE CONTROL GAP, NAMED

In the process as drawn, the front office originates the decision, obtains an operational sign-off, transmits the order, and then **verifies its own execution**. Operations can confirm that a trade settled; it cannot opine on whether deferring a loss was the right commercial judgement, whether the embedded financing rate is fair, or whether the cumulative deferred balance is within appetite. There is no independent risk approval, no independent price verification of the off-market rate, and no cumulative limit anywhere in the diagram.

This is not a hypothetical concern. It is the specific control weakness that every supervisory pronouncement on this technique — and the one large loss event associated with it — is addressed to. Section 15 sets out that record; section 18 sets out what the process should look like instead.

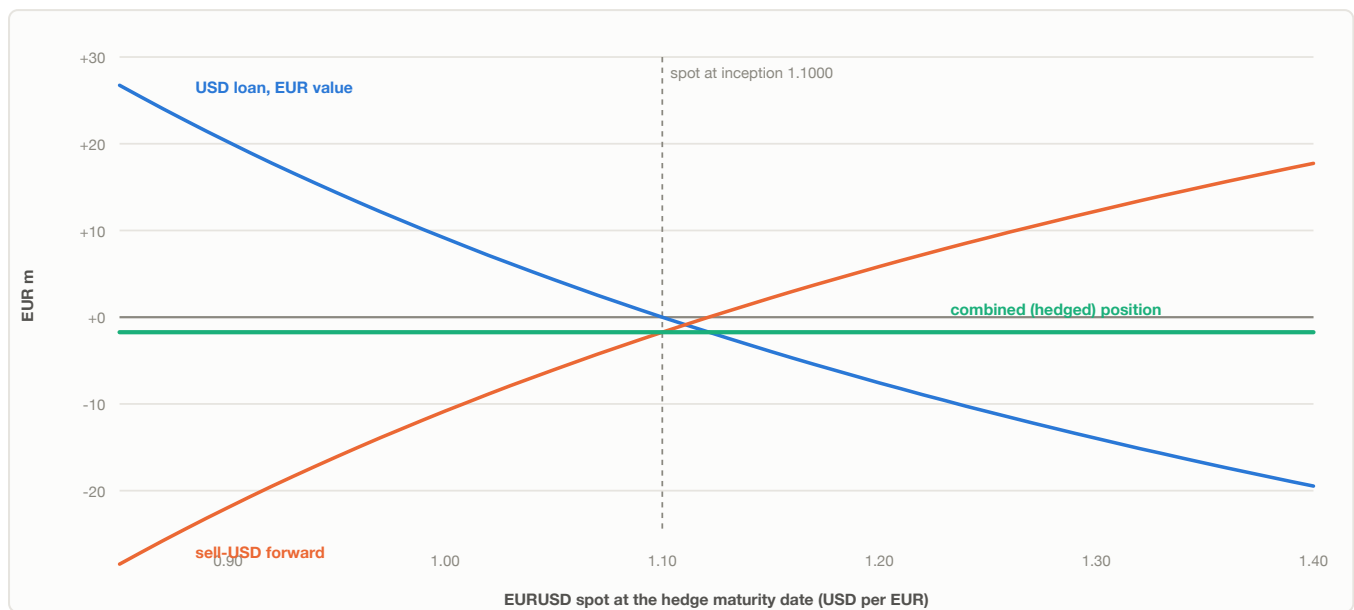
6 The payoff diagram — and why it is the wrong picture

You asked for a payoff diagram. Here it is, and it is worth being clear that it is *correct* and *almost entirely beside the point*.

FIGURE 1

The hedge is linear and it works: economic P&L is flat in the spot rate

USD 100m loan hedged with a one-year sell-USD forward struck at 1.1215



The combined line sits slightly below zero across the whole range: that gap is the forward premium the EUR investor pays away to hedge a higher-yielding currency, roughly the 2.0% USD-EUR rate differential. Nothing in this picture is controversial. Every question about historic rate rollovers concerns *when the cash moves*, not where this line sits.

An FX forward is linear. It has no optionality, no convexity, no gamma, no strike-dependent kink. Its payoff is a straight line in the spot rate, and when you add it to a foreign-currency asset the two lines cancel to a flat one. That flat line is the correct picture of a hedge and it is *identical* under a standard rolling hedge and under a historic rate rollover. A payoff diagram cannot distinguish between them, because they have the same payoff.

The difference between these two techniques is invisible on a payoff diagram, because a payoff diagram plots value against price and the entire difference is in time. Look for the difference in the cash-flow diagram (Figure 3) and the exposure diagram (Figure 6) instead.

This is worth internalising because it explains why the technique is easy to approve. Present it to a committee that thinks in payoff diagrams and Greeks and it registers as a no-op: same exposure, same delta, same VaR, no optionality added, and a helpful operational simplification thrown in. Every standard market-risk lens returns "no change". The risk this technique creates does not live in the market-risk lens at all — it lives in the counterparty credit lens, the liquidity lens, and the conduct lens, and it will not appear unless someone deliberately looks there.

IF YOU TAKE ONE THING FROM PART I

A historic rate rollover changes nothing about your market risk, your delta, your VaR, or your hedge effectiveness in the economic sense. It changes *when the money moves*, and it does that by creating an obligation to your dealer that did not previously exist. Every subsequent question in this document is about that obligation.

PART II

Why it is effective

The technique is genuinely effective at what it does. Being precise about what that is — and what it is not — is the difference between approving it with the right conditions and approving it with the wrong ones.

7 The conservation law

Compare the two techniques on a like-for-like basis. The standard hedge pays cash across five dates; the rollover pays once at the end. To compare them honestly you have to future-value the standard hedge's payments to the same terminal date at the fund's cost of money.

TABLE 4 · LIKE-FOR-LIKE COMPARISON, ALL VALUES AT YEAR FIVE

SCENARIO	STANDARD HEDGE, FUTURE- VALUED TO YEAR 5	ROLLOVER, TERMINAL SETTLEMENT	DIFFERENCE	AS % OF NOTIONAL
A · USD strengthens throughout	−27.58m	−28.04m	−0.46m	0.46%
B · FX round-trip, ends flat	−10.12m	−10.45m	−0.32m	0.32%
C · EUR strengthens throughout	+8.91m	+9.10m	+0.19m	—

EUR m. Standard-hedge cash flows compounded at EUR 2.5%. Scenario C shows the rollover ahead only because the model applies a symmetric accrual rate; in practice the dealer charges a spread when you owe and pays sub-market when you are owed, so the real-world Scenario C figure is also negative. See section 10.

THE CONSERVATION LAW

To within a financing spread of 30–50 basis points, **the two techniques produce the same total euro outcome.** The rollover does not reduce the FX loss. It does not improve the hedge. It does not lower the cost of hedging. It relocates cash flows through time, and charges you for the relocation. Anyone who describes it as reducing FX risk has misunderstood it, and that misunderstanding is the most common way this technique gets approved for the wrong reasons.

8 The decomposition: hedge plus unsecured loan

The cleanest way to hold this in your head — and the way to explain it to a board — is that a rolled position is exactly two things bolted together:

TABLE 5 · DECOMPOSITION OF A ROLLED FORWARD

COMPONENT	WHAT IT IS	WHO BEARS IT	WHERE IT SHOULD APPEAR
A market-rate rolling hedge	Exactly the position the fund would have had anyway: short USD forward, re-struck at market each period.	Fund's investors, as intended	Market risk: FX delta, VaR, hedge-effectiveness testing. Unchanged.
A drawn, unsecured, revolving loan from the dealer	Principal equal to the cumulative unsettled hedge loss. Interest at the accrual rate embedded in the far-leg strike. Repayable on demand, in the sense that the dealer can decline to roll.	Fund's investors, unintentionally — and the dealer, as credit risk	Counterparty credit exposure, fund leverage, liquidity ladder, borrowing disclosure. Usually appears in none of them.

The second row is the whole story. The strike arithmetic in section 4 is just the packaging: it takes a loan and expresses it as a forward rate so that it settles through the FX plumbing rather than through the treasury function. The economics are identical to the fund borrowing EUR 6.07m unsecured from the dealer at EUR + 75bp and using it to settle the hedge.

Test the claim by asking what a credit officer at the dealer does when this trade is booked. The answer, in a well-run bank, is that they book a credit line drawing. The Foreign Exchange Committee of the New York Fed made exactly this point in the guidance it has maintained on this subject since 1991: rollovers at historic rates "*virtually always involve the extension of credit by one party to the other,*" and unrealised losses so rolled "*should be booked as a credit extension to the client against established credit lines and subject to the market participant's normal assessment of the client's creditworthiness.*"

If your dealer's credit department is treating this as a loan to your fund, and your own risk function is treating it as an FX hedge, then only one of you is measuring the position correctly.

9 The three conditions that make it work at all

The technique delivers real value, but only where three conditions hold simultaneously. Each is a genuine constraint and each fails in identifiable situations.

Condition 1 — The trade must be uncollateralised

If the fund posts daily variation margin under a CSA, the mark-to-market on the hedge is being paid in cash every day regardless of what happens at the roll date. Deferring the settlement changes nothing; the cash has already gone out as margin. **The liquidity benefit of a historic rate rollover is exactly zero on a fully margin-ed book.**

The reason it is available at all to a private credit fund is a specific regulatory carve-out. Under the EMIR margin RTS, the requirement to exchange variation margin on physically-settled FX forwards and FX swaps applies only where both counterparties are credit institutions or MiFID investment firms. An AIF trading physically-settled FX forwards with a bank is outside that requirement. UK EMIR is aligned.

VERIFY THIS, DO NOT ASSUME IT

The regulatory carve-out permits the absence of margin; it does not create it. Your dealer's CSA is a commercial document and may well require variation margin on FX forwards regardless, or impose a threshold that this strategy would consume. **Read the CSA before the mechanism is assumed to work**, and check whether the trades are within an existing netting set or booked separately. If VM is contractually required, this entire proposal delivers no liquidity benefit and should be declined on that ground alone.

Condition 2 — The dealer must be willing to extend the credit

Because the trade is uncollateralised, the dealer is taking unsecured exposure to the fund that grows with every adverse roll. This consumes the dealer's credit line, attracts a CVA charge, consumes RWA, and is reviewed by their credit committee on a schedule the fund does not control. The dealer's appetite is the binding constraint on how much liquidity relief the technique can ever provide.

Condition 3 — The terminal cash flow must actually arrive

The deferral has to end somewhere. The technique creates value only if the accumulated balance settles on a date when the fund receives a matching cash inflow in a matching currency. If the asset repays on schedule, this works beautifully. If the asset prepays early, extends, restructures, or defaults, the deferred balance crystallises against an inflow that is early, late, smaller or absent. Section 21 is essentially an assessment of how confidently each private credit strategy can meet this condition.

WHERE IT GENUINELY EARNS ITS PLACE

Strip away the noise and the honest case is this: a fund holding **long-dated, contractually amortising, low-default, non-prepayable assets** with **predictable repayment dates** and a **modest hedge ratio relative to NAV** can use this technique to avoid holding a permanent cash buffer that would otherwise cost its investors 40–80bp a year in drag, in exchange for a financing spread of 30–50bp on a balance that is usually small. That is a real, defensible trade. It is also a much narrower claim than "this is how private credit hedges FX".

10 What it costs, in basis points

The cost is the dealer's spread over the accrual benchmark, applied to the deferred balance, for as long as it is outstanding. It is invisible unless you decompose the far-leg rate, because it arrives as a forward rate rather than an invoice.

TABLE 6 · COST OF DEFERRAL BY DEALER SPREAD — SCENARIO A, FIVE YEARS

DEALER SPREAD OVER EUR	TERMINAL BALANCE (EUR M)	COST VS STANDARD HEDGE (EUR M)	COST IN BP OF HEDGED NOTIONAL
50 bp	27.88	0.31	31 bp
75 bp	28.04	0.46	46 bp
100 bp	28.19	0.62	62 bp
150 bp	28.51	0.93	93 bp
200 bp	28.83	1.25	125 bp
300 bp	29.47	1.90	190 bp

Cumulative, over the five-year life, on a USD 100m hedge under a persistent 15% dollar rally.

Two points about this table. First, an unsecured, uncommitted, multi-year revolving facility to a levered credit fund is not a 50bp product. If the fund were to arrange the equivalent borrowing in the loan market it would pay 200–350bp. The dealer knows this. The question is whether they price it that way inside the forward points, and whether anyone at the fund would notice if they did.

Second, and more importantly: nobody is checking. A forward strike of 1.2158 against a market forward of 0.9889 is not a number that can be validated by eye, and the embedded financing rate cannot be extracted from it without solving the recursion backwards. Unless the fund independently reprices every rolled trade and reports the implied spread, the dealer's charge is unobservable.

A BEST-EXECUTION PROBLEM, NOT JUST A COST PROBLEM

An AIFM owes its investors a best-execution duty. A rolled forward is negotiated bilaterally, at a rate that by construction cannot be compared to a screen price, with a financing component that is not separately quoted. The only way to discharge the duty is to **decompose every roll into its market-rate forward and its implied financing rate, and benchmark the financing rate against an independent funding curve**. If that calculation is not in the operating model, the fund cannot demonstrate best execution on these trades, and the cost to investors is by definition undisclosed. Requiring the dealer to quote the roll as an explicit market-rate forward *plus* a separately stated financing rate solves this at essentially no cost, and any dealer who resists that request is telling you something.

The CRO lens

What a second-line function should conclude, what it should measure, and what it should insist on before this goes live. This is the part of the document to argue about.

11 The risk transformation map

The single most useful thing a risk function can say about this technique is that it is a *transformation*, not a mitigation. Risk does not leave the fund. It changes category, and in doing so it moves out of the reporting lines that were built to catch it.

TABLE 7 · WHAT THE TECHNIQUE CONVERTS, AND WHERE THE NEW RISK LANDS

RISK BEFORE	RISK AFTER	WHY THE CHANGE MATTERS TO SECOND LINE
Recurring, small liquidity demands — a payment every roll date, sized by one period's FX move	One large terminal liquidity demand — sized by the cumulative move over the asset's whole life, plus accrued interest	Frequent small demands are absorbed by ordinary treasury operations and are self-limiting: a bad quarter is followed by a fresh start. A single terminal demand is a cliff, is unbounded in size, and lands on one date the fund does not choose.
Settlement risk on a two-day cycle , bounded and CLS-netted	Uncollateralised counterparty credit exposure that compounds across the asset's life	This is the core conversion. A market risk you were being paid to run becomes a credit risk you are paying to run, concentrated in one or two dealers, uncollateralised and unlimited by anything except the dealer's own appetite.
Market risk, visible in VaR and FX delta	Rollover / funding risk , visible in nothing	Market risk is continuous and measurable. Rollover risk is binary and discontinuous: the dealer either rolls or does not. No VaR model produces that number, and it correlates with exactly the conditions that made the balance large.
Realised P&L, marked and settled each period	Deferred P&L carried inside an off-market strike	Valuation risk and investor-equity risk. If the off-market leg is not independently priced, NAV is wrong and the error accumulates in one direction. See section 14.
Trades executed at observable market rates	Trades executed at bilaterally negotiated off-market rates	Conduct risk and best execution. This is the specific practice that supervisors have singled out for thirty-five years, for reasons set out in section 15.
No embedded borrowing	Undisclosed, unsecured leverage ranking alongside other unsecured creditors	Leverage measurement, LP disclosure, and any borrowing covenant in the LPA or the subscription facility. The fund may be in breach of its own documents without anyone computing it.

The technique is not dangerous because it is complicated. It is dangerous because it is simple, it looks like nothing on a market-risk report, and every risk it creates sits in a different department's inbox.

12 Counterparty credit risk: sizing the line you actually need

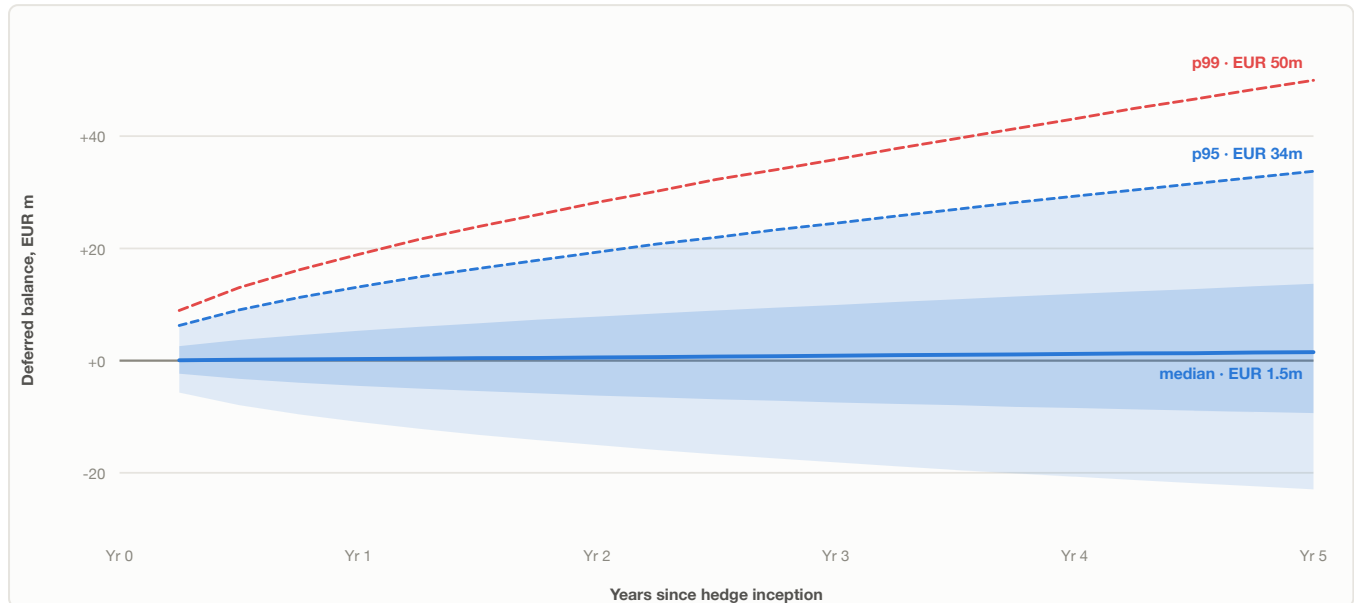
If the deferred balance is a loan, the first question is how big it gets. This is answerable. Model EURUSD as a log-normal process at 8% annualised volatility — roughly its long-run realised level — roll quarterly for five years,

and run the recursion from section 4 across 200,000 paths.

FIGURE 6

Distribution of the deferred balance: a benign median and a very fat tail

200,000 paths · EURUSD lognormal at 8% vol · quarterly rolls · USD 100m hedged



The median outcome is a balance near zero, which is exactly why this technique is easy to approve and hard to govern. The risk lives in the tail: the balance peaks above EUR 37m on 5% of paths and above EUR 53m on 1% of them.

TABLE 8 · DEFERRED BALANCE DISTRIBUTION, EUR M ON A USD 100M HEDGE

HORIZON	MEAN	MEDIAN	75TH	95TH	99TH	99.5TH
1 year	0.6	0.3	5.3	13.1	18.9	21.1
2 years	1.2	0.6	7.8	19.3	28.2	31.7
3 years	1.7	0.9	9.9	24.5	35.9	40.3
5 years	3.0	1.5	13.7	33.8	50.0	56.6
Peak over the life	12.6	—	—	37.3	52.7	—

Read the distribution before reading the tail. The median deferred balance after five years is EUR 1.5m on a USD 100m hedge — a rounding error. This is precisely why the technique is easy to approve: on the great majority of paths it does exactly what it says on the tin, quietly, for years. The peak balance exceeds 10% of notional on 49% of paths and 20% of notional on 24% of paths, and at the 99th percentile it reaches 53% of the hedged notional.

FIGURE 7

How big an unsecured credit line does this actually require?

Same model, varying FX volatility and hedge tenor

EURUSD vol	1-year hedge	3-year hedge	5-year hedge	7-year hedge
6%	10%	19%	27%	33%
8%	14%	27%	37%	47%
10%	18%	35%	49%	62%
12%	22%	44%	63%	80%

Peak deferred balance at the 95th percentile, as a percentage of the hedged notional.
Read as: the size of uncollateralised credit line the strategy needs to survive 19 years out of 20.

THE NUMBER THAT SHOULD DECIDE THE DEBATE

To run this strategy on a five-year asset at 8% FX volatility and survive at the 95th percentile, the fund needs an **uncollateralised credit line from its FX dealers equal to roughly 37% of the hedged notional**. At the 99th percentile it needs 53%.

No dealer extends a private credit fund an unsecured, uncommitted line of that size for FX. Typical unsecured FX lines for a mid-size fund run to single-digit millions to low tens of millions. Which means the honest conclusion is this: **the structure will fail before it needs to**. The binding constraint is not whether the fund can afford the eventual loss — it can, the loss is economically offset by the asset. The binding constraint is that the dealer's line runs out first, at which point the deferral stops and the fund must pay everything it has deferred, immediately.

Three refinements that make the credit picture worse rather than better:

- **Concentration.** A private credit fund runs FX with two or three dealers, not fifteen. The deferred balance concentrates into the same names, alongside whatever else the fund does with them. Section 19 proposes per-dealer limits for this reason.
- **Wrong-way risk, in both directions.** Sustained dollar strength is the classic global risk-off signal. It coincides with credit spread widening, bank funding stress and rising default rates in the fund's own book. The deferred balance is therefore largest exactly when the dealer's appetite is smallest and the fund's assets are performing worst. When the balance runs the other way — the dealer owing the fund — the fund holds an uncollateralised receivable from a bank in a stressed market, which is the exposure that vaporised in 2008 and was being actively repriced in March 2023.
- **The asymmetry of the accrual rate.** The model above accrues at a single rate in both directions. Real dealers do not. Expect to pay a credit spread over the benchmark when you owe, and to receive something below the benchmark when you are owed. Over a full cycle in which the currency round-trips, that bid-offer is a pure, permanent leak.

13 Liquidity risk: the cliff and the line-pull

The technique is sold as a liquidity management tool, and it does improve liquidity — in the ordinary state of the world. What it does to the tail is the opposite, and the tail is second line's job.

The line-pull

Suppose that at the year-three roll date, with EUR 18.28m deferred, the dealer's credit committee declines to extend further. Nothing has broken; a credit officer has simply done their job. The fund must now settle the maturing contract in cash.

TABLE 9 · THE LINE-PULL, SCENARIO A AT YEAR THREE

MEASURE	ROLLOVER STRATEGY	STANDARD ROLLING HEDGE
Cash required at the year-three roll date	EUR 18.28m	EUR 5.01m
Notice period	Two business days	Two business days
Was the demand foreseeable from the previous quarter's reporting?	Only if the deferred balance was being reported	Yes — it is the period's FX move
Amplification of the liquidity event	3.7×	1.0×

The amplification factor grows with time. By year five it would be 6.6×. This is the defining property of the technique from a liquidity standpoint: **it does not remove the liquidity event, it stores up several of them and delivers them as one, at a moment chosen by the counterparty.**

The structural ceiling

There is also a hard mathematical limit, visible in the strike equation from section 4. The new strike is $K = N \div [N/F - D(1+c\tau)]$. As the deferred balance D approaches the euro proceeds of an at-market leg, the denominator approaches zero, the required strike goes to infinity, and past that point the deferred loss cannot be expressed as a forward rate at all. In Scenario A the fund reaches 27% of that ceiling by year five. Under a 30% dollar rally it would be past 60%. The structure has a breaking point that is not a policy choice and cannot be negotiated.

FIGURE 5

By year five the booked rate is 38% away from the market — visible to anyone who looks

Scenario A · the widening wedge is the accumulated, unsettled loss



This is the single most useful surveillance signal in the whole structure. A rolled position carries a strike that no market-maker would quote, and the distance from market is a direct, unambiguous read-out of how much unsettled loss the fund is carrying. Any exception report that flags forward strikes more than a set distance from the prevailing curve will find every rolled trade, every time.

The interaction with fund-level liquidity

For an open-ended or semi-liquid vehicle — an evergreen private credit fund, an ELTIF 2.0, a LTAF, an interval fund — the deferred balance is a senior claim that sits ahead of redeeming investors and is not visible in the liquidity ladder unless someone puts it there. It should be modelled as a contingent call on cash at every horizon bucket, and it should be included in reverse stress testing: the question "what combination of FX move and redemption rate breaks this fund" has a materially different answer once a deferred balance is running.

The over-hedge trap

Private credit assets do not repay when the model says they will. Direct lending loans on five-to-seven year paper have historically shown average lives of two to three years because sponsors refinance. When the asset disappears, the hedge does not. The fund is left with an outright short-dollar forward position — a naked FX bet, prohibited by most investment policies — and to unwind it, it must crystallise the deferred balance. The rollover therefore makes an over-hedge **expensive to correct at exactly the moment correcting it becomes urgent**, and creates a live incentive to leave the over-hedge running. That incentive is itself a risk.

14 Valuation, NAV and investor equity

This section matters more than its length suggests, because it is where a mechanical issue becomes a fiduciary one.

The far leg of a rolled swap is off-market by construction. Its fair value is not zero. In Scenario A at the year-three roll, the fund books a forward at 1.2158 when the market forward is 0.9889: the contract has a fair value of roughly **negative EUR 18.3m** at the moment of execution. If the valuation process carries newly executed forwards at par on the assumption that a new trade is worth nothing — which is the correct assumption for every other forward the fund executes — the NAV is overstated by that amount.

TABLE 10 · NAV OVERSTATEMENT IF THE OFF-MARKET LEG IS CARRIED AT PAR

YEAR	UNMARKED DEFERRED BALANCE (EUR M)	AS % OF A EUR 250M NAV	CONSEQUENCE
1	6.07	2.43%	Within most materiality thresholds. Nobody notices.
2	12.85	5.14%	Above materiality. A redeeming investor exits at a 5% premium funded by those who stay.
3	18.28	7.31%	An error of this size found late is a restatement and a regulatory notification.
4	23.01	9.20%	Performance fees have been accrued and possibly paid on an overstated NAV.
5	28.04	11.22%	Terminal correction lands in a single reporting period.

The investor-equity point is the sharp one. An overstated NAV is not a neutral accounting error; it is a transfer. Every investor who redeems while the deferred balance is unmarked is paid out with money belonging to the investors who remain. In an evergreen vehicle with quarterly liquidity, that transfer runs continuously and compounds. In a closed-end fund it shows up as performance fees accrued on gains that were never real.

SHARE-CLASS HEDGING MAKES THIS MATERIALLY WORSE

If the technique is applied to *share-class* hedging rather than portfolio hedging, the deferred balance belongs to the holders of a specific share class at the time it was incurred. Roll it forward and it is borne by whoever holds that class later. Investors who subscribe into a hedged share class carrying an unmarked deferred loss are buying a liability they cannot see, at a price that does not reflect it. This is a hard cross-investor equity problem with no clean answer, and my recommendation in section 27 is that the technique should not be used for share-class hedging at all.

Hedge accounting

Under IFRS 9, a derivative designated as a hedging instrument with a non-zero fair value at designation is permitted, but the off-market element is a financing component and must be identified and accounted for as such. It is not part of the hedge. Practically this means: the rolled forward must be split into its at-market hedge and its financing element; only the former can be designated; the latter runs through P&L as financing cost; and hedge effectiveness must be tested against a hypothetical derivative struck at market, not against the booked strike. If the auditors have not been consulted on this before the first roll, they will be after it, and the answer will be more expensive then.

15 Conduct risk: the regulatory record and why it exists

This is the section to read before deciding that the technique is a routine market convention. It is a convention, but it is a convention that supervisors have written about specifically, repeatedly, and for a reason.

The reason is mechanical. A historic rate rollover has exactly one economic property that no other FX technique has: it converts a realised loss into an unrealised one, at the discretion of the person doing the trade, without any cash moving. Every legitimate use of the technique relies on that property. So does every illegitimate one.

TABLE 11 · THE SUPERVISORY RECORD

SOURCE	STATUS	WHAT IT SAYS
FX Global Code, Principle 12	Current, global, and adhered to by every major FX dealer	"Transactions should be conducted at prices or rates based on the prevailing market conditions at the time of the transaction. Exceptions to this, such as historical rate rollovers, should be covered by internal compliance policies. " The technique is named explicitly, in the Execution principles, as an exception requiring a compliance policy.
NY Fed Foreign Exchange Committee, guidance maintained since 1991 and reissued in 1995 under the title <i>"Historical Rate Rollovers: A Dangerous Practice"</i>	Long-stand- ing market guidance	Rollovers at historic rates "virtually always involve the extension of credit by one party to the other"; they can be used "to conceal losses, evade taxes, or defraud his or another trading institution"; unrealised losses so rolled should be booked as a credit extension against established lines. Non-market rates should not be permitted in interbank dealing and elsewhere only "with strict management oversight".
APRA, Prudential Standard APS 180 para- graph 17	Binding on Australian ADIs	Historic rate rollovers are permitted only on FX contracts; they are prohibited on all non-FX transactions. Non-compliant rollovers must be reported to the supervisor. The concern is stated as pricing integrity — off-market restructuring can conceal losses or effect improper risk transfers.
Allied Irish Banks / Allfirst, USD 691m loss, 2002; Ludwig Report	The refer- ence loss event	Historic rate rollovers were a central mechanism by which a single trader deferred and concealed accumulating FX losses over five years. The Ludwig Report's finding was not that the technique was exotic but that the control environment around it was weak: the trader originated the rollovers, and no one independently questioned strikes that were visibly far from market.

Take the AIB case seriously without over-reading it. Nothing in the proposal in front of you resembles fraud. But the reason that case is the standing reference is that it demonstrated how ordinary the concealment looks from the outside. There was no exotic instrument. There were forwards, rolled, at strikes that did not match the market, booked by a person whose own performance was measured on the P&L those strikes determined, with an operational sign-off that confirmed the trades had settled and no independent challenge to whether they should have been done. The Federal Reserve had been warning about the technique for eleven years before it happened.

THE UNCOMFORTABLE STRUCTURAL POINT

In the process as proposed, the portfolio manager decides at each roll date whether to defer a loss. That is a discretionary decision, made by the person whose track record the deferred loss would otherwise appear in, with no independent party required to agree. It does not matter that the individuals are trustworthy. The control question is not whether they will misuse the discretion; it is whether the firm would **know** if they did. As drawn, the answer is no — and that is what a supervisor will ask, in those words.

The surveillance answer, and it is a good one

There is a straightforward and complete detective control, visible in Figure 5. Every rolled trade carries a strike that no market-maker would quote, and the distance from market is a direct read-out of the unsettled loss. A daily exception report that flags any FX forward struck more than a defined distance from the prevailing curve will catch every rolled trade, every time, with no false negatives. Reconciling the sum of those distances to an independently computed deferred balance closes the loop. This is cheap, it is mechanical, and its absence is inexcusable.

16 Accounting, leverage and regulatory reporting

TABLE 12 · REPORTING AND REGULATORY TOUCHPOINTS

REGIME	QUESTION TO RESOLVE BEFORE GO-LIVE
AIFMD leverage (gross and commitment methods)	The forward notional is captured either way. The <i>embedded borrowing</i> is not automatically captured by either method, yet it is unambiguously leverage in substance. Does it need to be added, and does adding it breach the leverage limit disclosed to investors in the Article 23 disclosure?
Annex IV reporting	Each roll creates two new trades. Turnover, counterparty exposure and borrowing fields all move. Confirm the deferred balance is reflected in the borrowing and counterparty-exposure fields rather than netting to zero.
LPA and facility covenants	Most LPAs cap indebtedness and most subscription lines carry a negative pledge or an indebtedness covenant. An unsecured obligation to an FX dealer, ranking pari passu with other unsecured creditors, is capable of tripping both. This needs a legal opinion, not a view.
EMIR / UK EMIR reporting	Every roll is a lifecycle event plus two new trades, reported T+1. Confirm the off-market rate is reported as the actual contract rate; confirm the near leg's early-termination character is correctly represented. Reporting rolled trades at market rates would be a reporting breach.
UMR / initial margin	Rolls inflate gross notional. Physically-settled FX forwards and swaps are excluded from the IM calculation itself, but confirm the AANA methodology and whether the roll volume interacts with any other in-scope product. If the fund is close to a threshold this matters.
IFRS 9 hedge accounting	Split the financing element; designate only the at-market hedge; test effectiveness against a hypothetical derivative struck at market. Agree the treatment with the auditors before the first roll, not after.
Tax	Some jurisdictions treat the closing of the maturing contract as a realisation event regardless of whether cash moved. For a Luxembourg or Irish fund platform this is usually manageable; for onshore feeders it may not be. The FXC's original concern about rollovers being used to shift income between reporting periods is a tax concern as much as a conduct one.
MiFID II best execution (at the AIFM)	See section 10. Without decomposition into a market forward plus a stated financing rate, best execution on these trades cannot be demonstrated.

17 Operational and settlement risk

- **The near leg must match to the character.** The entire benefit depends on the near leg matching the maturing trade exactly — notional, currency pair, value date, direction, rate. A mismatch of a single day or a single unit means cash actually moves, on a gross basis, in both currencies. On a USD 100m leg that is a USD 100m unfunded payment obligation arriving without warning. Pre-settlement matching against the maturing trade must be a hard system control, not a checklist item.
- **CLS netting is an assumption, not a fact.** The mechanism assumes both legs are eligible, submitted and matched in the same CLS session with the same counterparty. Verify eligibility for every currency pair in scope (several emerging-market pairs are not CLS-eligible), and specify the fallback when a leg fails to match: the resulting exposure is gross principal for value that day, which is Herstatt risk of the classic kind.
- **Reason codes are a governance artefact.** The proposed use of a specific order reason code is correct and important, but only if the code is mandatory, cannot be overridden by the front office, feeds the exception report automatically, and is reconciled to the independently computed deferred balance. A reason code that exists only to explain an odd rate to the dealing desk is decoration.
- **Confirmations and books-and-records.** Confirmations will carry off-market rates. Any confirmation matching tolerance built around "rate within X of market" will generate persistent breaks. Fixing that by

widening the tolerance would disable the firm's principal detective control for mis-booked FX — the tolerance should instead be bypassed only for trades carrying the reason code, and every such bypass should be logged.

- **Custodian and administrator.** The administrator must be able to price an off-market forward correctly, and must not be netting it to zero on execution. Get written confirmation of the pricing methodology for off-market forwards from the administrator before go-live. This is the single most common place where the NAV problem in section 14 actually originates.
- **Key-person and continuity.** The deferred balance is knowledge held in a small number of heads. If the portfolio manager leaves mid-life, does the successor understand that the 1.2158 strike on the blotter is not a market rate? The deferred balance should be a named, reported, reconciled quantity in the fund's own books, independent of the trade file.

18 The control framework: fourteen requirements

If the technique is approved, this is the minimum. Each item exists because of a specific failure mode identified earlier in this document; the cross-reference is given so the list can be defended rather than merely imposed.

TABLE 13 · CONTROL REQUIREMENTS

#	CONTROL	REQUIREMENT	ADDRESSES
1	Independent approval per roll	Every rollover requires prior sign-off from a named second-line individual with authority to refuse, in addition to operations. The front office cannot self-approve. Approval is per roll, not a standing permission.	§5, §15
2	Documented commercial rationale	Each roll records, at the time, why deferral is preferred to settlement: the specific liquidity constraint, the expected source of the terminal cash flow, and its date. "Market convention" is not a rationale.	§9, §15
3	Cumulative deferred balance limit	A hard notional cap, per dealer and in aggregate, expressed in currency and as a percentage of NAV. Breach forces settlement, not escalation. See section 21.	§12, §13
4	Treated as a credit line drawing	The deferred balance is booked as unsecured exposure to the dealer, consumes a credit limit approved by the credit function, and appears on the counterparty exposure report daily — not netted, not offset against the hedge.	§8, §12
5	Independent price verification of the off-market rate	Every rolled trade is decomposed into (a) the market forward and (b) the implied financing rate. The financing rate is benchmarked to an independent funding curve. Deviations beyond tolerance are escalated.	§10, §14
6	Off-market strike exception report	Daily automated report of every FX forward struck more than a defined distance from the prevailing curve, reconciled to the deferred balance. Reviewed by second line, not the desk.	§15
7	Correct NAV treatment, confirmed in writing	The off-market leg is marked at fair value from trade date. Written confirmation from the administrator of the pricing methodology for off-market forwards, obtained before go-live.	§14, §17
8	Mandatory, non-overridable reason code	System-enforced at order entry; feeds the exception report automatically; cannot be applied or removed by the front office after execution.	§17
9	Near-leg matching as a hard system control	Automated pre-settlement match of the near leg against the maturing trade on all five fields. Failure blocks the trade rather than raising an alert.	§17
10	Stress testing in the fund liquidity framework	The deferred balance under a 15% and a 25% adverse FX move, with and without a dealer line-pull, at every liquidity horizon bucket. Included in reverse stress testing.	§13
11	Explicit line-pull contingency	A documented, pre-agreed funding plan for settling the full deferred balance within two business days at any point in the asset's life: named sources, sized, tested annually.	§13
12	Asset-hedge linkage register	Each rolled hedge is mapped to the specific asset funding its terminal settlement, with that asset's expected repayment date. Reviewed at every roll. Amendment, extension, prepayment or default of the asset triggers mandatory review of the hedge.	§9, §13, §24
13	Legal opinion on indebtedness	Written opinion on whether the deferred balance constitutes indebtedness for the purposes of the LPA, the subscription facility and AIFMD leverage disclosure.	§16
14	Investor disclosure	Disclosure in the offering documents and periodic reporting that the fund defers FX settlement through off-market rollovers, that this constitutes unsecured borrowing from FX counterparties, and the current outstanding balance.	§14, §16

THE THREE THAT ARE NON-NEGOTIABLE

If the programme has to start with a subset, start with **#1 (independent approval)**, **#3 (a hard cumulative cap)** and **#7 (correct NAV treatment)**. The first prevents the conduct failure, the second bounds the credit and liquidity failure, and the third prevents the fiduciary failure. Controls 5, 6 and 12 should follow within the first quarter. Everything else is a matter of sequencing.

19 Sizing the limit: what the deferred balance is made of

A natural proposal is to cap the exposure at the interest rate differential plus the funding cost of the rollovers — on the reasoning that if the position is fully hedged, that is all that can remain, whatever spot does. The economics behind that reasoning are right. The limit that falls out of it is not, and the gap between the two is the most important thing in this section.

The identity

The deferred balance decomposes into exactly three terms, with no residual:

DECOMPOSITION OF THE DEFERRED BALANCE

Each period's loss against the market forward splits cleanly:

$$N/S_t - N/F_{t-1} = (N/S_{t-1} - N/F_{t-1}) + (N/S_t - N/S_{t-1})$$

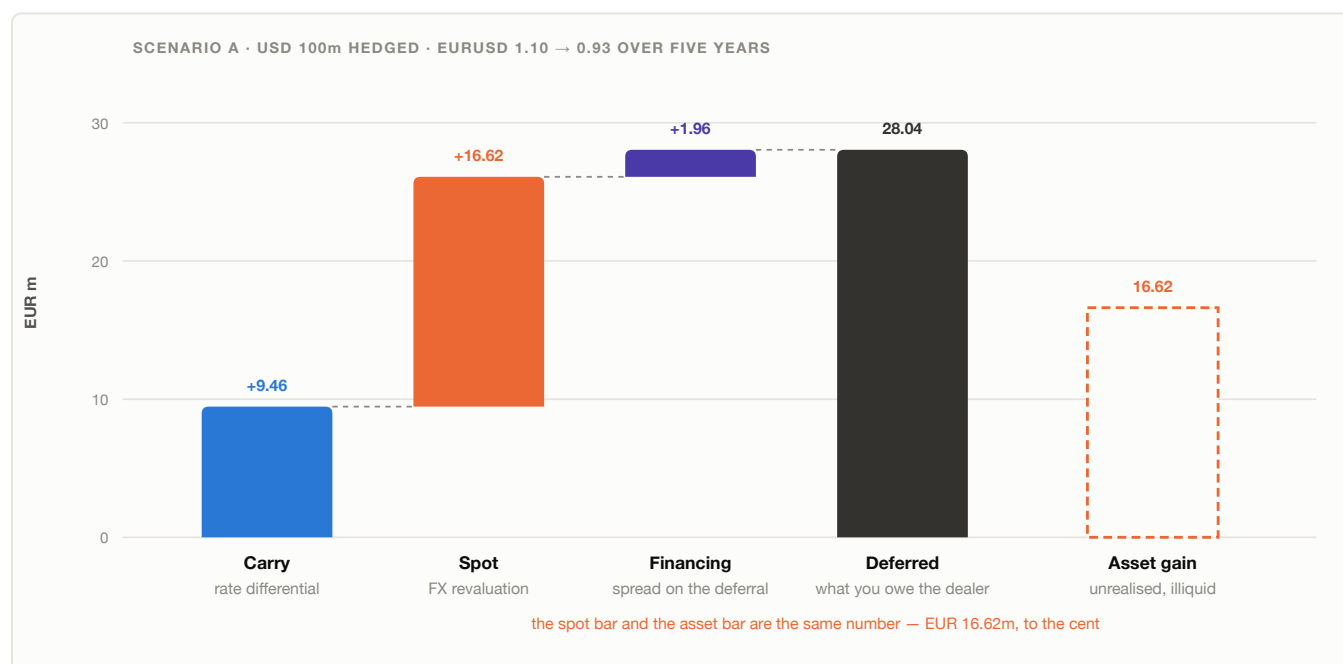
total loss = **carry** (what you lose if spot never moves) + **spot** (the FX move itself)

So $D_T = \text{carry} + \text{spot} + \text{financing}$, and the cumulative **spot** term is identically equal to the base-currency revaluation of the foreign-currency asset — the unrealised gain sitting inside the loan.

FIGURE 8

What the deferred balance is actually made of

The two blue-and-violet bars are cost. The orange bar is not — it is your asset's unrealised gain, and it is the biggest.



A limit built on “rate differential plus funding cost” prices the first and third bars and ignores the second. That second bar is unbounded, it is what the dealer lends you, and it settles in cash. The offset is genuine, but it is locked inside a loan the dealer cannot net against and the fund cannot spend.

The proposed formula prices the first and third bars and omits the second. That omission is not carelessness: the spot term genuinely is offset, which is exactly why it feels safe to leave out. But it is the largest of the three, it is unbounded, and it is what the dealer lends you and what settles in cash.

TABLE 14 · THE IDENTITY ACROSS ALL THREE SCENARIOS

SCENARIO	CARRY	SPOT	FINANCING	DEFERRED BALANCE	ASSET REVALUATION
A · USD strengthens throughout	9.46	16.62	1.96	28.04	16.62
B · FX round-trip, ends flat	9.10	0.00	1.35	10.45	0.00
C · EUR strengthens throughout	7.95	−16.28	−0.76	−9.10	−16.28

EUR m on a USD 100m hedge over five years. The spot column and the asset-revaluation column are equal to the cent on every path, and the three components sum to the balance with zero residual — these are identities, not approximations.

Scenario B is the instructive one. The currency round-trips and finishes exactly where it started, so the spot term is **precisely zero** and the proposed formula holds perfectly. It is the one case in which the reasoning is fully correct — and even there the answer is EUR 10.45m, **10.5% of notional, more than double the proposed 5%**.

Why the offset does not help you

Three reasons, each independent, each sufficient on its own:

1. **The dealer nets only what is in the netting set.** Your ISDA nets your derivatives against each other. The loan is not in it. The dealer's exposure is the gross amount you owe on the forward, and their credit department sizes the line accordingly — regardless of how well hedged you believe you are.
2. **The offset is illiquid and cannot fund the settlement.** The asset gained EUR 16.62m and you cannot spend a cent of it. The settlement is in cash, in two days, and the offsetting gain does not become cash until the loan repays — which in Scenario A is four years after the first roll.
3. **The offset can disappear while the hedge survives.** On a default or a prepayment the asset is gone; the deferred balance and the forward are not. The netting you were relying on was never contractual — it was an assumption about the asset still being there.

The proposed formula measures net economic cost. A limit has to measure the gross unsettled obligation. The two differ by the entire unrealised gain trapped inside an illiquid loan.

Even on its own terms, 5% is too small

Set the spot term to zero — grant the assumption entirely — and compute what carry alone accumulates to. This is the proposed formula, evaluated correctly:

TABLE 15 · DEFERRED BALANCE WITH SPOT PERFECTLY FLAT, AS % OF HEDGED NOTIONAL

HEDGE TENOR	DIFFERENTIAL 1.0%	1.5%	2.0%	2.5%
1 year	1%	1%	2%	2%
3 years	3%	4%	5%	7%
5 years	5%	7%	9%	12%
7 years	7%	10%	13%	17%

Includes the 75bp financing spread on the accumulating balance.

At today's USD–EUR differential, a five-year hedge accumulates 9% of notional in carry before the currency moves a single pip. A 5% figure corresponds to roughly a three-year hedge at a 2% differential, or a five-year hedge at 1%. It is not a general number, and it is tenor-dependent in a way the formula conceals.

Where 5% is exactly the right number

Now the useful part. Run the deferred balance through 120,000 paths and ask how often each candidate cap would actually bind:

FIGURE 9

Calibrating the cap: how often would each level actually bind?

A 5% cap on a five-year hedge is hit on 68% of paths — median time to first breach, four rolls



This is how a limit level gets defended rather than asserted. Read down a column to pick the percentile you want the limit to bind at; read across a row to see how the same number behaves as tenor extends. The 5% row is highlighted because it is the level under discussion.

A 5% cap on a five-year hedge is breached on **68% of paths, with a median time to first breach of one year — four rolls**. Thirty-six per cent breach inside the first year.

THE REFRAME

As a *risk-appetite ceiling*, 5% is a limit that spends most of its life in breach, and a limit permanently in breach is one that gets waived, then ignored, then removed. As a **mandatory reset trigger** — "settle the balance to zero in cash whenever it exceeds 5% of the leg" — it is arguably the single best rule available, because it confines the technique to short-term smoothing and structurally prevents it from becoming a multi-year deferral.

Same number, opposite consequence on breach. Keep the 5%; write it as a trigger, not a ceiling, and pair it with a genuine ceiling further out. That is what section 21 does.

20 Pricing the roll: checking the strike against market data

Control 5 requires independent price verification of the off-market rate. This section is that control, written out as arithmetic you can run from a terminal.

The difficulty is that a rolled strike cannot be sanity-checked by eye. A forward booked at 1.2158 against a market of 0.9889 looks like a typo, and there is no screen price to compare it to. But the strike is not arbitrary — it is a deterministic function of the market forward and the deferred balance, so the process can be run backwards to recover what the dealer has charged.

THE CHECK, IN FIVE STEPS

Inputs — two numbers off the screen, three off the tickets: market spot S ; market forward points for the far-leg tenor; the maturing contract's all-in rate K_{old} ; the new far leg's all-in rate K_{new} ; the notional N and tenor τ .

1. Market forward: $F = S + \text{points}/10,000$
2. Owed at the roll date: $D_{roll} = N/S - N/K_{old}$ (this is just the mark-to-market of the maturing leg)
3. Owed at far-leg maturity, as embedded in the new strike: $D_{mat} = N/F - N/K_{new}$
4. Implied financing rate: $c = (D_{mat} / D_{roll} - 1) / \tau$
5. Compare c to your own funding benchmark, and reconcile D_{roll} to the deferred balance in the register. Both must agree.

TABLE 16 · THE CHECK APPLIED TO EVERY ROLL IN SCENARIO A

ROLL	SPOT	FWD POINTS	K_{OLD}	K_{NEW}	D AT ROLL	D AT MATURITY	IMPLIED RATE	IMPLIED SPREAD
Year 1	1.0500	204.9	1.1215	1.1475	6.07	6.27	3.250%	75.0 bp
Year 2	1.0000	195.1	1.1475	1.1790	12.85	13.27	3.250%	75.0 bp
Year 3	0.9700	189.3	1.1790	1.2158	18.28	18.87	3.250%	75.0 bp
Year 4	0.9500	185.4	1.2158	1.2580	23.01	23.76	3.250%	75.0 bp

EUR m. The model was built with a 75bp spread over EUR 2.5%; the check recovers 75.0bp at every roll from screen data alone. Nothing about the dealer's charge needs to be taken on trust.

USE THE TRADED FORWARD POINTS, NEVER A FORWARD DERIVED FROM RATE CURVES

It is tempting to build F from covered interest parity using the two deposit or OIS curves. Do not. Cross-currency basis means CIP does not hold, and the error goes straight into the implied spread:

BASIS IGNORED	FORWARD USED	IMPLIED SPREAD	ERROR
0 bp (correct)	1.0705	75 bp	—
10 bp	1.0694	229 bp	+154 bp
20 bp	1.0683	383 bp	+308 bp
30 bp	1.0673	538 bp	+463 bp

A 20bp basis error — unremarkable for EURUSD, and much larger over year-end — misstates the dealer's charge by more than 300bp and would generate a false escalation every quarter. The forward points already embed the basis; the rate curves do not. Pull the points, not the curves.

Running it on real data

On a terminal the inputs are the spot rate and the outright forward or forward points for the far-leg value date — the FX forward calculator and the FX forwards curve page both produce them, and the forward points are also available as individual tenor tickers. Two practical notes. First, match the **value date**, not the tenor label: a rolled far leg rarely lands on a clean 3M or 1Y date, and interpolating between standard tenors is fine but must be

documented. Second, use a **fixed daily snap** — a published benchmark fixing rather than a live mid — so that the check is reproducible and the desk cannot dispute the timestamp.

Three outputs are worth recording every time, because together they are the whole control:

- **D at roll**, reconciled against the deferred-balance register. A break means a missing or mis-booked trade.
- **The implied spread**, against the Layer 1 limit. This is the cost the investor is paying and it exists nowhere else in the fund's records.
- **The strike distance**, against the Layer 0 limit. This is the number the exception report should have flagged independently — if it did not, the exception report is broken.

21 The limit architecture: four layers

One limit cannot carry this. The technique creates risk in three different units — a cost measured in basis points of yield, an obligation measured in currency, and a hedge integrity measured in notional — and a limit expressed in one of those units is structurally blind to the other two. The architecture below separates them, and adds a fourth layer that needs no model at all.

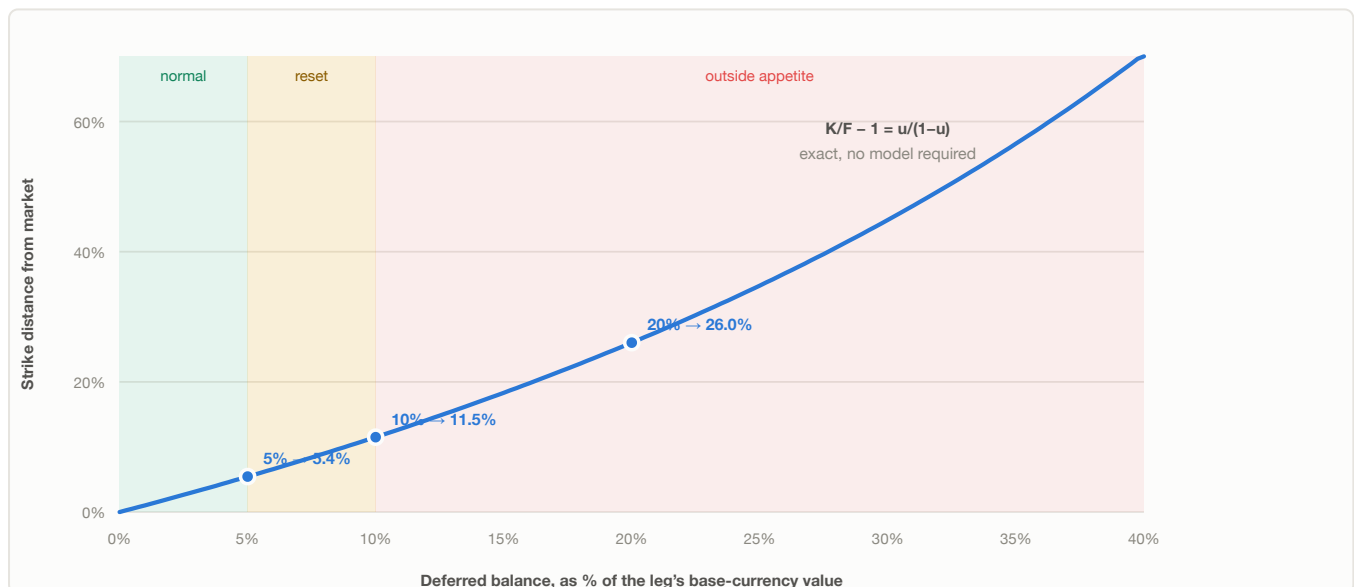
Layer 0 — the master observable

Start here, because it is the only limit in the framework that can be enforced by someone with a screen and no model, register or reconciliation. The strike arithmetic gives an exact one-to-one relationship between how far a booked strike sits from the market and how much deferred loss it contains:

FIGURE 10

The limit you can enforce from a screen

Strike distance is a one-to-one function of the deferred balance — so capping one caps the other



Verified against every roll in Table 2: at year three the identity predicts a booked strike of 1.2158 against a market forward of 0.9889, which is exactly what the model produces. A dealing-desk rule reading “no forward may be booked more than 5.4% from the prevailing curve” is a 5% deferred-balance limit that needs no register, no model and no reconciliation to enforce.

TABLE 17 · LAYER 0

LIMIT	LEVEL	BASIS AND CONSEQUENCE
Strike distance from market on any single booked forward	5.4% soft 11.5% hard	Exactly equivalent to a 5% / 10% deferred-balance limit via $K/F-1 = u/(1-u)$, verified against every roll in Table 2. Enforceable at the point of execution by the dealing desk, and independently by an overnight exception report. Hard breach blocks the trade.

The virtue of this limit is that it cannot be gamed by a reporting failure. Every other limit here depends on the deferred balance being correctly computed and reported somewhere; this one is visible on the ticket.

Layer 1 — cost limits: what is hedging costing the investor?

This is where the "rate differential plus funding cost" formula belongs. It is the right way to measure the cost of the programme; it is simply not a measure of exposure.

TABLE 18 · LAYER 1

LIMIT	LEVEL	BASIS AND CONSEQUENCE
Annual hedge carry cost as % of gross portfolio yield	25%	At a 2% differential on an 8.5–9.5% gross yield the programme already consumes 21–24% of gross spread, so this limit is close to binding to-day. That is worth knowing before anyone treats hedging as free. Breach triggers a hedge-ratio review, not a settlement.
Implied financing spread on any rolled trade, per section 20	+150 bp soft +250 bp hard	Benchmarked to an independent funding curve. This is the only place the dealer's charge is visible. Hard breach means the roll is not executed without treasury sign-off.
Cumulative cost of deferral YTD, bp of hedged notional	Report	Monitored and disclosed rather than limited. Answers "what did this convention cost us this year?" — a question that otherwise has no answer anywhere in the fund's records.

Layer 2 — exposure limits: what do we owe, and can we pay it?

Two units are needed here, and this is where the proposal to net the forwards against NAV belongs. It is the right instinct, and stronger than it might appear.

THE MARK-TO-MARKET OF THE HEDGE BOOK IS THE DEFERRED BALANCE

At every roll date, the mark-to-market of a contract struck at K and valued at spot S is $N/K - N/S$ — and that equals the deferred balance identically:

ROLL	BOOKED STRIKE	SPOT	MTM OF THE BOOK	DEFERRED BALANCE
Year 1	1.1215	1.0500	–6.07	–6.07
Year 3	1.1790	0.9700	–18.28	–18.28
Year 5	1.2580	0.9300	–28.04	–28.04

So a limit on **aggregate mark-to-market of the FX hedge book as a percentage of NAV** is the deferred-balance limit, expressed in a number the administrator already produces every valuation cycle. No register, no model, no separate calculation. **Its one dependency is control 7** — if the off-market leg is carried at par on execution, the MTM reads zero and the limit is blind. The limit and the valuation control stand or fall together.

One calibration warning. A NAV-denominated limit is the right unit for the fund-solvency question, but its tightness depends entirely on how much of the fund is hedged:

TABLE 19 · WHAT A NAV-BASED CAP IMPLIES, BY HEDGE INTENSITY

HEDGED NOTIONAL AS % OF NAV	3% NAV CAP = % OF NOTIONAL	BREACH FREQUENCY, 5Y	5% NAV CAP = % OF NOTIONAL	BREACH FREQUENCY, 5Y
25%	12.0%	43%	20.0%	24%
50%	6.0%	64%	10.0%	49%
75%	4.0%	72%	6.7%	61%
100%	3.0%	76%	5.0%	68%

The same NAV limit is comfortable for a lightly hedged fund and permanently binding for a fully hedged one. Pair it with a per-hedge limit in percent of notional, which is invariant to fund size.

TABLE 20 · LAYER 2

LIMIT	LEVEL	BASIS AND CONSEQUENCE
Aggregate MTM of the FX hedge book as % of NAV, absolute value	2% soft 3% hard	The headline exposure limit. Sized so a full crystallisation is absorbable from ordinary liquidity without a capital call. Uses administrator data. Hard breach forces settlement of the excess at the next roll.
Deferred balance per hedge % of the leg's base-ccy value	5% reset 10% hard	The 5% is a <i>mandatory reset trigger</i> , not a ceiling — settle to zero and re-strike at market. It binds on 68% of five-year paths by design, which is the point. The 10% hard cap sits near the median peak and keeps every position clear of the structural ceiling.
Per-dealer deferred balance	1.5% of NAV or the approved unsecured line, lower of	Concentration. Must sit inside a credit limit approved for that legal entity alongside all other exposure to it.
Structural ceiling utilisation $u = D(1+ct) \div (N/F)$ — the fraction of the leg consumed	15% soft 20% hard	Past this the strike arithmetic degrades sharply and unwinding becomes disorderly. A mathematical limit, not a preference — see section 13.
Stressed MTM 25% instantaneous adverse move	8% of NAV	Soft. Breach triggers a documented reduction plan to the risk committee.
Maximum consecutive rolls	4	Forces a periodic return to a market rate and a fresh commercial decision. Prevents indefinite deferral becoming the default.
Maximum deferral tenor	Lesser of 3 years and the linked asset's expected repayment	Ties the deferral to the cash flow meant to extinguish it.
Crystallisation concentration % of aggregate balance settling in any one month	40%	Prevents the deferred balances across the book from stacking onto a single date and recreating the cliff at portfolio level.

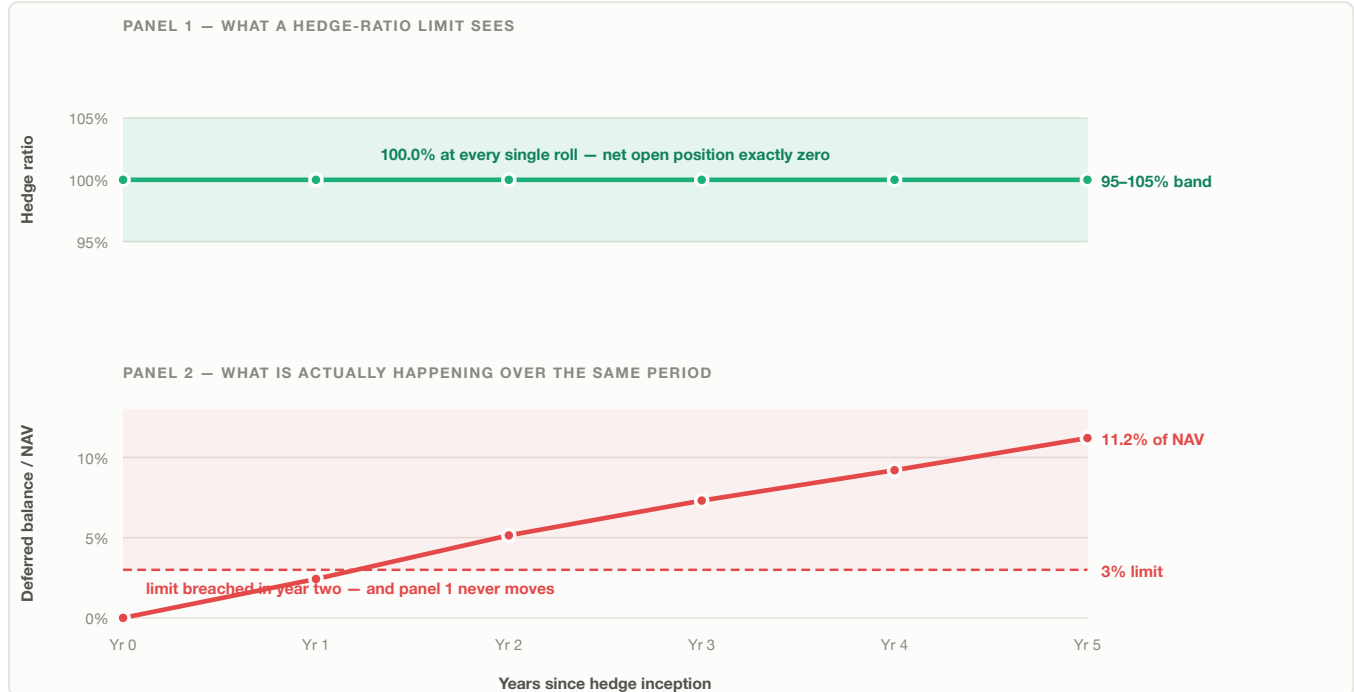
Layer 3 — hedge-integrity limits: is this still a hedge?

Hedge ratio and net open position are the correct way to answer that question, and they belong in any FX framework whether or not rollovers are used. But it is important to be precise about what they will and will not catch.

FIGURE 11

Why a hedge-ratio limit cannot see this risk

Same fund, same five years, same trades — Scenario A on a EUR 250m NAV



Hedge ratio and net open position are computed on *notional*; the deferred balance is a *valuation* quantity. The hedge is doing its job perfectly throughout — that is not the failure mode. A hedge-ratio limit is necessary and it is the right way to answer “am I still hedged?”, but it will return a clean reading every single month while the deferred balance runs to four times its limit. The two limits are orthogonal and neither substitutes for the other.

A HEDGE-RATIO LIMIT IS BLIND TO THIS RISK BY CONSTRUCTION

Throughout Scenario A the hedge ratio is 100.0% and the net open position is exactly zero at every roll, while the deferred balance climbs to 11.2% of NAV. That is not a modelling artefact — it is the definition. Hedge ratio and net open position are computed on **notional**; the deferred balance is a **valuation** quantity. They are orthogonal. Both are necessary; neither substitutes for the other, and a framework that has only the hedge-ratio limit will report a clean position every month for five years.

TABLE 21 · LAYER 3

LIMIT	LEVEL	BASIS AND CONSEQUENCE
Hedge ratio, per currency	95–105%	Hedge notional over the base-currency value of the assets in that currency. The core "am I still hedged" test. Rebalance at the next roll on breach.
Net open position, per currency	2% of NAV	Absolute residual exposure. Catches the case where the ratio looks fine on a small currency but the euro amount is material.
Aggregate net open position	5% of NAV	Across all currencies, without netting long against short unless the pairs are genuinely correlated.
Notional drift vs linked asset	5% soft 10% hard	Per asset, not just per currency. Catches amortisation, PIK accrual and partial prepayment before they aggregate into a currency-level breach.
Minimum residual asset life at roll	12 months	Prevents deferring into an asset about to disappear, which converts the hedge into an outright position.
Hedge effectiveness vs a hypothetical derivative struck at market	90–110%	Tested against the at-market hedge, never against the booked strike — otherwise the financing element contaminates the test. Required for IFRS 9 in any case.
Notional rolling in any single week	25%	Roll-date laddering. Prevents the whole book maturing on one date and turning a roll refusal into a single fund-level event.
Historic-rate rolls as % of all rolls	30%	The exception-density monitor. Above this the technique has stopped being an exception, which is the specific condition FX Global Code Principle 12 is addressed to. Escalate to the risk committee.

WHEN TWO LAYERS CONFLICT, HEDGE INTEGRITY WINS

The layers can pull against each other. When an asset partially prepays, Layer 3 says reduce the hedge notional; reducing it crystallises part of the deferred balance, which is precisely what Layer 2's reset trigger exists to make happen but which the desk will resist because it converts an unrealised loss into a realised one. **Write the tie-break into the policy now, before it is live: rebalance and take the cash.** An unhedged position is a breach of investment policy; a deferred balance is a breach of a risk limit, and a policy that lets the second one justify the first has the hierarchy inverted.

THE REGISTER

All twenty limits ship as a populatable workbook, `HRR_Limit_Register.xlsx`, alongside this document: one row per limit with its definition, both thresholds, the breach direction, frequency, data source, owner and consequence, plus a computed red/amber/green status. It carries the red-flag checklist from section 22, the calibration grids behind every level, and a live version of the section 20 strike check into which market data can be pasted directly. It is generated from the same model functions as this document, so the two cannot drift apart.

22 Surveillance: fifteen red flags

Limits catch magnitude. They do not catch pattern, and every documented failure of this technique was a pattern that no single limit would have breached. Each flag below states the signal, the mechanism that produces it, and what to look at.

TABLE 22 · RED FLAGS, MOST SEVERE FIRST

#	SIGNAL	WHY IT HAPPENS — AND WHY IT MATTERS	OBSERVABLE
1	A roll on a hedge whose asset has prepaid, defaulted or been written down	The hedge no longer hedges anything. It is now an outright FX position that is also being financed by the dealer, and the cash flow that was supposed to extinguish the balance no longer exists. This is the clearest single flag in the set, and it is the Allfirst fact pattern: positions rolled with nothing underneath them.	Asset-hedge linkage register vs the roll blotter. Any roll where the linked asset's status is not "performing".
2	Strike distance widening across three or more consecutive rolls	The recursion $D_t = D_{t-1}(1+c\tau) + FX$ compounds whenever the FX term keeps its sign. Monotonic widening means a trending currency and no reset — the one path on which the structure fails rather than merely costs money.	Sign of the change in $ K/F-1 $ over the last three rolls, per position.
3	Historic-rate rolls rising as a share of all rolls	The Code treats this as an exception requiring a compliance policy. An exception applied to most rolls is not an exception; it is the operating model, and the per-roll commercial-rationale test has stopped being applied even if the forms are still being signed.	Layer 3 exception-density ratio, trended over 12 months.
4	Rolls on short-dated, self-liquidating assets	Trade finance and drawn revolvers generate cash inside the settlement window, so no liquidity mismatch exists. Deferral without a liquidity justification has no commercial rationale at all — only a P&L one.	Roll blotter filtered by strategy and by linked-asset tenor under 12 months.
5	Deferred balance growing while spot is flat or favourable	Arithmetically the balance can then only be growing from carry plus financing. If it is growing faster than carry explains, either the dealer's spread is punitive or a trade has been mis-booked. There is no third explanation.	Change in balance vs the carry-only projection from Table 15.
6	Roll requests clustering near period ends	The technique's defining property is converting a realised loss into an unrealised one. Liquidity needs are indifferent to quarter-ends; P&L optics are not. Clustering is the signature of the second motive.	Histogram of roll request dates against the reporting calendar.
7	A near leg that does not match the maturing trade on all five fields	If notional, pair, value date, direction and rate do not all mirror, the legs do not net — cash moves gross, and the difference is a new position established inside what looks like a roll. This is the mechanism by which additional risk gets added without a new trade decision.	Pre-settlement match report; any tolerance override.
8	Deferred balance not reconciling to the sum of strike distances	These agree by construction — $u/(1-u)$ is exact. A break means a trade is missing from the register, a strike is mis-booked, or a roll happened outside the process.	Daily reconciliation of the register against the position file.
9	Deferred balance rising while reported hedge P&L is flat	The loss is being carried in the strike instead of the P&L. This is the NAV failure in section 14 happening in real time, and it is the point at which an accounting problem becomes an investor-equity problem.	Hedge P&L in the NAV pack vs the change in the deferred balance.
10	Rolls migrating toward a single dealer	Rolls require a dealer willing to extend unsecured credit. If only one still is, you have simultaneously discovered your capacity constraint and maximised your concentration — and an unusually accommodating dealer is recovering it somewhere in the points.	Share of rolled notional by counterparty, trended.
11	Implied financing spread widening	It is the dealer's own read on your credit and on how much of their line you have consumed. It typically moves weeks before their credit committee says anything formally, which makes it the earliest available warning of a line-pull.	Section 20 check, trended per dealer.
12	Hedge notional not stepping down with amortisation	Creates a growing over-hedge. Under this technique correcting it costs cash, so it persists — the structure penalises the correct action, and the position drifts from hedge toward position.	Layer 3 notional-drift limit, per asset.
13	Balance exceeding the linked asset's remaining expected cash flows	The deferral has no exit. Even a fully performing asset can no longer extinguish it, so the balance must be funded from somewhere else — and nobody has been asked where.	Register: balance vs remaining scheduled principal and interest.

#	SIGNAL	WHY IT HAPPENS — AND WHY IT MATTERS	OBSERVABLE
14	Rising confirmation breaks or tolerance overrides on rolled trades	Off-market rates break rate-tolerance matching, and the natural fix is to widen the tolerance — which disables the firm's principal detective control for mis-booked FX across the whole book, not just these trades.	Override log, filtered to the roll reason code.
15	The same person requesting and approving a roll	The entire risk is discretionary loss deferral by the person whose record the loss would otherwise appear in. The question is never whether they would misuse it, but whether the firm would know.	Approval log: requester and approver identity, per roll.

AND ONE THAT IS NOT A METRIC

Watch for the **reasoning** red flag: any limit, paper or committee answer that justifies the size of the deferred balance by pointing at the offsetting gain on the asset. It is the single most natural mistake in this subject — the offset is real, it is just not available, not in the netting set, and not guaranteed to survive. When you hear it, the person saying it is measuring net economic cost and calling it exposure. Section 19 exists to be handed to them.

Suitability for private credit

The source deck lists six strategies in a single table under one shared comment. Those six have materially different cash-flow profiles, and the technique's suitability differs from "yes, with conditions" to "no" across them. Applying it as a platform-wide convention is the largest single risk in the proposal.

23 The five suitability gates

The technique works when the deferred balance has a known, reliable, matching cash flow to die into, and when the liquidity problem it solves is real. Five questions establish both. A strategy that fails any one of them should not use it.

TABLE 23 · THE GATES

#	GATE	WHY IT IS A GATE
1	Is the repayment date knowable and stable?	The deferred balance settles when the hedge is finally closed. If that date is at the borrower's option, or depends on a third party's behaviour, the deferral has no reliable terminus and the fund is deferring into uncertainty.
2	Is the asset genuinely illiquid over the roll horizon?	The whole justification is that the fund cannot generate cash from the asset within the settlement window. If the asset self-liquidates every 90 days, there is no problem to solve, and deferring a loss without a liquidity justification is exactly the pattern the FX Global Code requires a policy for.
3	Is the notional stable?	A rolled hedge is a fixed notional. If the asset's outstanding balance moves at the borrower's discretion, the fund is perpetually over- or under-hedged, and under this technique correcting an over-hedge requires crystallising the deferred balance — which creates an incentive not to correct it.
4	Is credit loss uncorrelated with the FX stress?	Sustained dollar strength is a global risk-off signal. If the strategy's defaults cluster in exactly those conditions, the deferred balance peaks in the same month the assets funding it fail. That is wrong-way risk at the portfolio level, and it is not diversifiable.
5	Is the hedge tenor long relative to the roll period?	The technique earns its keep by suppressing many settlements. One or two rolls do not justify the credit, governance and conduct overhead. But too many rolls, over too long a life, let the balance compound past any sensible limit.

24 Strategy by strategy: six verdicts

TABLE 24 · GATE ASSESSMENT

STRATEGY	1 DATE	2 ILLIQUID	3 NOTIONAL	4 NO WWR	5 TENOR	VERDICT
Infrastructure Debt	✓	✓	✓	✓	△	SUITABLE, CAPPED
Direct Lending	△	✓	△	△	✓	CONDITIONAL
NAV Lending	✗	✓	✓	△	✓	MARGINAL
Venture Debt	✗	✓	△	✗	△	NOT SUITABLE
Revolving Credit Financing	✗	△	✗	△	✗	NOT SUITABLE
Trade & Working Capital	✓	✗	✗	△	✗	NOT NEEDED

✓ passes · △ passes with qualification · ✗ fails

Infrastructure Debt — THE BEST FIT, WITH ONE HARD CONSTRAINT

This is the strategy the technique was designed for, whether or not anyone designed it. Infrastructure debt is long-dated, contractually amortising against a defined schedule, prepayment-locked or heavily penalised, secured on assets with regulated or contracted revenues, and carries default rates a fraction of corporate credit. Repayment dates are knowable years in advance and the notional steps down predictably. Gates 1–4 all pass cleanly.

The constraint is gate 5, and it cuts the other way from everywhere else. A 20-year infrastructure loan implies eighty quarterly rolls. Over that horizon the deferred balance has time to compound to a size no dealer will carry, and the strike drifts so far from market that the position becomes operationally unmanageable. The answer is a **mandatory reset to a market rate at a fixed interval** — settle the balance in cash every two or three years, funded from scheduled amortisation, and start again. That preserves the liquidity benefit within each window while capping the credit exposure. It also has the merit of forcing a fresh commercial decision on a schedule rather than at the dealer's discretion.

Direct Lending — CONDITIONAL, AND THE CONDITION IS PREPAYMENT

Direct lending passes on illiquidity and tenor: the assets genuinely cannot be sold at price inside a settlement window, and five-to-seven year paper hedged quarterly means twenty-plus rolls. The liquidity problem is real and the technique addresses it.

The problem is gate 1. Direct lending loans do not run to maturity. Sponsors refinance, and the historical average life on five-to-seven year paper has run at two to three years. The prepayment is at the borrower's option, it is not forecastable at the loan level, and it clusters — refinancing windows open and shut for the whole market at once. So the terminal cash flow that is supposed to extinguish the deferred balance arrives on a date the fund does not control and cannot predict.

In fairness, early prepayment is the benign version: the cash arrives, the balance settles, and the fund is whole. The genuinely bad case is the other one:

DEFAULT WITH A LIVE DEFERRED BALANCE

Scenario A, year three. EURUSD has fallen to 0.97, the deferred balance is EUR 18.28m, and the borrower defaults with a 40% expected recovery.

Recovery proceeds, USD 40m at 0.97	EUR 41.24m
Deferred FX balance now due to the dealer	EUR -18.28m
Residual hedge on a notional that no longer exists	USD 60m short, outright
Net euro proceeds on EUR 90.9m deployed	EUR 22.96m

The technique did not cause the credit loss and did not increase it — under a standard hedge the fund would have paid the same FX cost, spread across three years and funded out of coupon income. What it did was **stack the entire FX settlement on top of the credit event, in cash, in the same week, funded from a recovery process that has not yet produced any money.** And it leaves a USD 60m outright short-dollar position live on the book, which under this same technique will not settle in cash either — it will compound into the same balance.

Direct lending can use the technique, but only with control 12 (the asset-hedge linkage register) operating properly, with mandatory review on any amendment, extension or covenant breach, and with the deferred balance sized so that a simultaneous default and crystallisation is absorbable.

NAV Lending — MARGINAL; SHORT TENORS ONLY

NAV facilities look superficially well-suited: two-to-four year bullets, secured on a diversified pool, low historical loss. But gate 1 fails in a specific and important way. Repayment depends on distributions from the underlying private equity funds — and the timing of those distributions is not merely uncertain, it is *systematically* uncertain in the direction that hurts. Distribution activity slows precisely when exit markets close, which is when the borrowing GP most wants to keep the facility outstanding. Extension is the norm, not the exception, and the 2022–24 distribution drought is the working example.

So the fund is deferring FX losses into a repayment date that is likely to move, in an environment where it is likely to move at the same time as the FX stress. If the technique is used here it should be on tenors no longer than two years, with the deferred balance settled in cash at any extension of the underlying facility, and with the extension case explicitly modelled rather than treated as remote.

Venture Debt — NOT SUITABLE

This one is not close, and it fails on the gate that matters most.

Venture debt runs default rates well above senior corporate credit, on borrowers whose collateral is enterprise value rather than hard assets — and enterprise value in a distressed venture-backed company does not decline gracefully, it evaporates when the next funding round fails to arrive. Repayment dates are nominal: the loans are short-dated on paper and routinely extended, restructured or repaid early on an exit. Notional is unstable because of tranching drawdowns and warrant exercises.

But the disqualifying issue is gate 4. Venture debt defaults cluster in risk-off environments — funding markets close, rounds do not get done, runway runs out. Those are the same environments that produce sustained dollar strength. The deferred FX balance and the credit losses in the book are driven by the same underlying factor, in the same direction, at the same time.

WRONG-WAY RISK, CONCRETELY

A risk-off quarter delivers, simultaneously: a 10% dollar rally that adds several points of notional to the deferred balance; a cluster of portfolio defaults where the collateral has already melted; a dealer credit committee reviewing its unsecured line to a venture debt fund in exactly that market; and no incoming cash from a portfolio whose repayments depend on exits that are not happening. Every one of those is the same event. There is no diversification anywhere in the structure, and the fund's own collateral-melt analysis already establishes that the recovery assumption fails in precisely this state of the world.

Revolving Credit Financing — NOT SUITABLE

Revolvers fail on notional stability, which is the gate no control can fix. The outstanding balance moves at the borrower's option, on short notice, in both directions. A rolled hedge is a fixed-notional instrument. The fund would be structurally over-hedged whenever the facility is repaid down — holding an outright short-dollar position — and under-hedged whenever it is drawn.

Under a standard rolling hedge that is a manageable nuisance: resize the hedge at the next roll and settle the difference in cash. Under this technique, resizing down means crystallising a share of the deferred balance, which creates a standing incentive to leave the over-hedge in place. A structure that penalises the correct risk-management action is badly designed regardless of how it is governed. Add that revolver drawdowns are typically 30–180 days — so the underlying does not need a multi-year hedge at all — and the case disappears entirely.

Trade & Working Capital — NOT NEEDED, AND THAT IS THE PROBLEM

Trade finance fails gate 2, which sounds like a technicality and is in fact the most serious finding in this section. The assets are 30–180 day self-liquidating receivables. The book turns over several times a year and generates cash continuously. There is no liquidity mismatch between the hedge and the asset, because the asset is a short-dated cash flow. The hedge tenor matches the asset tenor and every settlement is funded by a receivable maturing in the same window.

Which means that if the technique is used here, it is being used to defer a loss with **no liquidity justification at all**. That is precisely the fact pattern that the FX Global Code's Principle 12 requires a compliance policy to address, that the Foreign Exchange Committee called dangerous, and that the AIB case turned on. It is also, in practice, the easiest place for it to happen: the rolls are frequent, the amounts individually small, and the operational machinery is already built. Including trade finance in the approved list of this technique creates the one genuine conduct exposure in the whole proposal, in exchange for no benefit whatsoever.

THE SINGLE MOST IMPORTANT RECOMMENDATION IN THIS DOCUMENT

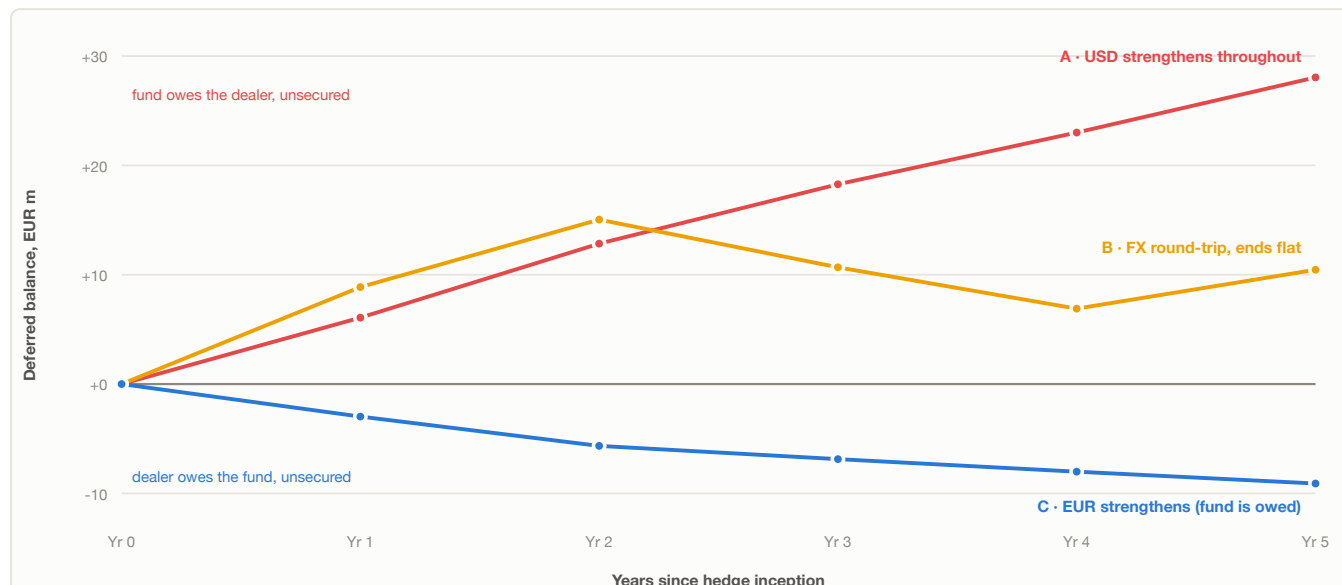
Do not approve this as a platform-wide convention. Approve it **strategy by strategy**, against the gates in Table 23, with the eligible list written into the limit schedule. The source deck's use-case table — six strategies, one shared comment, no differentiation — is the risk, not the technique.

25 Scenario library: where it works and where it breaks

FIGURE 4

The deferred balance is a credit exposure that ratchets — in whichever direction the currency trends

Balance accrues at EUR 2.5% + 75bp credit spread



Scenario B is the case people have in mind when they call the technique benign: the currency round-trips and the balance comes back down. Note that even there it never returns to zero — five years of accrued spread and one 8-figure excursion leave EUR 10.5m outstanding on a currency pair that finished exactly where it started.

TABLE 25 · SCENARIOS WHERE THE TECHNIQUE WORKS

SCENARIO	WHAT HAPPENS	OUTCOME
Range-bound currency EURUSD oscillates in a 5% band	The deferred balance moves up and down around a small mean and never reaches a level anyone needs to think about. The fund avoids twenty small settlement events and the associated cash buffer.	WORKS AS INTENDED Balance under 1% of NAV throughout; saves 40–80bp of cash drag.
Currency round-trip Scenario B: 1.10 → 0.98 → 1.10	The balance builds to EUR 15m at year two and falls back as the currency recovers. The fund never had to fund the excursion.	WORKS, WITH A CAVEAT Balance is still EUR 10.5m at year five on a pair that finished flat — accrued spread does not round-trip.
Amortising infrastructure loan, favourable currency	Balance runs in the fund's favour. Scheduled amortisation steps the notional down predictably. The dealer owes the fund a modest, declining amount.	WORKS But note the exposure has reversed: the fund now holds an uncollateralised receivable from a bank.
Loan repays on schedule, currency moved against	Year-five repayment of USD 100m arrives on the same day the EUR 28m deferred balance settles. Matched cash flow, matched currency, matched date.	WORKS EXACTLY AS DESIGNED This is the case the technique is built for.

TABLE 26 · SCENARIOS WHERE IT BREAKS

SCENARIO	WHAT HAPPENS	OUTCOME
Persistent trend A 20–25% move over 2–3 years, as in 2014–15 or 2021–22	The balance compounds monotonically. There is no mechanism that reduces it and no natural stopping point. It consumes the dealer line, then the limit, then the structure.	FAILS Balance reaches 20–30% of notional. Forced crystallisation.
Dealer withdraws the line	A credit committee declines to extend. The full accumulated balance becomes payable in two business days.	FAILS 3.7× the equivalent standard-hedge demand at year three; 6.6× by year five. Happens when markets are worst.
Borrower defaults with a balance running	FX settlement stacks on top of the credit event. A live over-hedge remains on a notional that no longer exists.	FAILS See the worked case in section 24. Two losses, one week.
Unexpected prepayment	The asset disappears; the hedge does not. Unwinding requires crystallising the balance, so the economically correct action carries an immediate cash cost.	DEGRADES Cash is available from the repayment, but the incentive now runs against correcting the over-hedge.
Redemptions in an ever-green vehicle	The balance is a senior claim ranking ahead of redeeming investors and absent from the liquidity ladder unless someone added it.	FAILS Redemption gate risk, plus the investor-equity transfer in section 14.
Correlated stress in venture debt	Dollar rally, default cluster, dealer line review, and no incoming cash — all driven by the same factor.	FAILS Wrong-way risk with no diversification anywhere in the structure.
Applied where there is no liquidity mismatch Trade finance, short revolvers	A loss is deferred with no commercial justification. Nothing goes wrong operationally, which is why it persists.	CONDUCT EXPOSURE This is the pattern the supervisory guidance exists to prevent.

26 The alternatives, and when to prefer them

The strongest argument against this technique is not that it is dangerous. It is that there is a better instrument for the same problem, and the comparison is not close.

TABLE 27 · ALTERNATIVES

ALTERNATIVE	HOW IT ADDRESSES THE SAME PROBLEM	WHEN TO PREFER IT
A committed FX liquidity facility	An explicit, documented, committed revolving credit facility sized to the stressed hedge settlement, drawn to fund settlements and repaid from asset cash flows. Economically identical to what the rollover does implicitly.	Almost always. See the box below — this is the recommendation.
Standard rolling hedge with a sized liquidity buffer	Hold cash or committed undrawn capacity equal to the stressed settlement. Costs 40–80bp of drag.	Where the hedge ratio is small relative to NAV and the drag is tolerable. The honest baseline against which everything else should be compared.
Reduce the hedge ratio	Hedge 70–80% rather than 100%. Cuts the cash demand proportionally and leaves a small deliberate FX exposure that partially offsets in the asset.	Where LPs will accept residual FX risk. Often the simplest real answer, and it costs nothing.
Bought options or a collar	A bought option caps the cash outflow at the premium, paid up front. Convex where the forward is linear, so the tail is bounded by construction. No credit extension, no off-market rate, no conduct question.	Where the concern is specifically the <i>tail</i> liquidity demand rather than the ordinary one. The premium is explicit and disclosed, which is a feature.
Cross-currency swap matched to amortisation	Exchanges cash flows on a schedule matched to the loan, removing the roll entirely.	Infrastructure debt with a contractual amortisation schedule. Eliminates rollover risk at the cost of a wider initial spread.
Natural hedge — fund the asset in its own currency	Borrow in USD at the fund or SPV level against USD assets. No FX derivative, no settlement, no basis.	Where the fund has leverage capacity. Structurally the cleanest solution and the most underused.
Currency-matched share classes	Offer a USD share class to USD-preferring investors instead of hedging their exposure back.	Where the LP base can be segmented. Removes the problem rather than financing it.

THE COMPARISON THAT SHOULD SETTLE IT

A historic rate rollover and a committed FX liquidity facility do the same economic thing: they let the fund defer paying an FX settlement by borrowing from a bank. The difference is entirely in the wrapper.

	HISTORIC RATE ROLLOVER	COMMITTED FX FACILITY
Commitment	Uncommitted — the dealer may decline any roll	Committed for the facility term
Pricing	Embedded in a forward rate; not separately quoted; not benchmarkable	An explicit margin over a benchmark
Credit approval	Implicit; often no formal drawing recorded on either side	Through both credit committees, sized and documented
Visibility in the fund's own reporting	None, unless deliberately constructed	A borrowing, on the balance sheet
Investor disclosure	Typically none	Disclosed as leverage
Behaviour in stress	Withdrawn precisely when needed	Available, subject to covenants
Conduct profile	Named in the FX Global Code as an exception requiring a compliance policy	An ordinary credit facility

Every line favours the facility. And there is a diagnostic buried in the comparison: if the fund asks its dealers for a committed facility of the size the model in section 12 says it needs, and they decline — that is the same credit decision they are making implicitly every time they agree to a roll, except made explicitly, in advance, by people whose job it is to make it. **A refusal is not a reason to fall back on the rollover. It is information about how much of this exposure the market is actually willing to fund.**

Decision

What I would sign, what I would not, and what I would ask.

27 Recommendation and conditions precedent

The technique is legitimate, it is genuinely used in the market, and there is a real problem it solves. It is also a named exception in the global conduct code for FX, the mechanism in the reference FX loss event of the last twenty-five years, and a structure whose tail requires more unsecured credit than any dealer will provide. Both of those statements are true, and a recommendation has to hold both.

Recommend: approve, narrowly and conditionally

1. **Approve for Infrastructure Debt and Direct Lending only**, with the reset and linkage conditions in section 24. Decline for Venture Debt, Revolving Credit Financing and Trade & Working Capital. Treat NAV Lending as marginal and approve only on tenors of two years or less, with extension of the underlying facility as a mandatory settlement trigger.
2. **Prohibit for share-class hedging** in all strategies, on cross-investor equity grounds (section 14).
3. **Make controls 1, 3 and 7 conditions precedent to the first trade** — independent per-roll approval, a hard cumulative cap, and written confirmation from the administrator on fair-value treatment of off-market forwards. Controls 5, 6 and 12 within the first quarter. The remainder within six months.
4. **Adopt the limit architecture in section 21**, recalibrated to actual fund sizes and actual dealer lines, with the aggregate deferred balance as a hard limit rather than a trigger.
5. **Require the parallel workstream on a committed FX facility** (section 26) to run alongside, and report back within two quarters. If a facility can be arranged on reasonable terms, it should replace this technique rather than supplement it.
6. **Report the deferred balance monthly** as a named line in the risk pack, with its history, dealer distribution, stressed value and days-to-crystallisation.

Would not recommend

- Approving it as a platform-wide FX convention across all six listed strategies.
- Approving it with operations sign-off as the only control gate.
- Approving it before the CSA has been read and the variation-margin position confirmed (section 9, condition 1). If VM is contractually required, the proposal delivers no benefit and should be declined outright.
- Approving it on the basis that peers use it. Peer usage establishes that the technique exists; it establishes nothing about the control environment inside those peers, and the reference loss event in this space happened at a firm whose peers were also doing it.

28 What would change my mind

Stating this explicitly is more useful than a verdict, because it tells the sponsor what evidence to bring.

TABLE 28 · EVIDENCE THAT WOULD MOVE THE ASSESSMENT

IF IT WERE ESTABLISHED THAT...	THEN
Dealers will provide committed uncollateralised capacity sized to the p95 peak balance, documented and irrevocable within the roll period	The line-pull scenario — the single most damaging failure mode — is removed, and the case strengthens materially across all strategies.
The far-leg financing rate is separately quoted and benchmarkable , with the dealer providing both the market forward and the financing rate on every roll	The best-execution and cost-transparency objections in section 10 largely fall away, and control 5 becomes trivial to operate.
The administrator confirms in writing that off-market forwards are marked at fair value from trade date , with an independent price source	The NAV and investor-equity objections in section 14 fall away. This is the cheapest of the three critical conditions to satisfy.
The strategies in scope are genuinely non-pre-payable with contractual amortisation — true infrastructure debt, not "direct lending with a hard call"	Gate 1 passes cleanly and the terminal settlement becomes reliable rather than hoped-for.
The proposed use is tactical — a small number of rolls to bridge a known, dated cash flow — rather than a standing convention	The credit accumulation problem largely disappears, and the control burden can be scaled down accordingly. This is a much easier approval.

29 Twenty questions for the approval meeting

Ordered so that the first five will settle most of it.

THE FIVE THAT MATTER MOST

- 1. Does our CSA with each FX dealer require variation margin on physically-settled FX forwards?**
If yes, this proposal delivers no liquidity benefit at all. Nothing else needs discussing until this is answered.
- 2. What is the maximum cumulative deferred balance, and who can say no?** If the answer is "we have not set one" or "the PM decides", the proposal is not ready.
- 3. Is the off-market forward marked at fair value from trade date, and has the administrator confirmed that in writing?**
- 4. What is each dealer's uncollateralised credit limit for us, when is it reviewed, and what happens to our fund if it is halved in a stressed quarter?**
- 5. Why not a committed FX liquidity facility instead?** If the answer is that dealers will not provide one at this size, ask why they are willing to provide the same credit implicitly.

CREDIT AND LIQUIDITY

6. Is the deferred balance recorded as a credit-line drawing against an approved limit, on both sides?
7. What is the deferred balance under an instantaneous 25% adverse FX move, as a percentage of NAV?
8. What is the funding plan if the entire balance becomes payable in two business days, and has it been tested?
9. How does the balance appear in the fund's liquidity ladder and reverse stress test?
10. What is our concentration to the single largest FX dealer, across all products, including this?

VALUATION, ACCOUNTING AND DISCLOSURE

11. Have the auditors confirmed the IFRS 9 treatment — financing element split out, effectiveness tested against a hypothetical derivative struck at market?
12. Does the deferred balance constitute indebtedness under the LPA and the subscription facility? Whose written opinion is that?

13. Is it captured in the AIFMD leverage calculation, and does capturing it breach the disclosed limit?
14. What do investors know? Is the technique disclosed in the offering documents and the outstanding balance reported?
15. Are performance fees being accrued on a NAV that includes an unmarked deferred loss?

CONDUCT AND OPERATIONS

16. Where is our internal compliance policy for off-market transactions, as required by FX Global Code Principle 12?
17. What exception report flags forward strikes materially away from market, who reads it, and does it reconcile to an independently computed deferred balance?
18. How do we demonstrate best execution on a bilaterally negotiated off-market rate?
19. Is the near-leg match against the maturing trade a hard system block or a manual check?
20. Which specific asset repays the deferred balance, on what date, and what happens to the hedge if that asset prepays, extends or defaults?

THE CLOSING THOUGHT

There is nothing wrong with a fund borrowing from its bank to manage a cash-flow mismatch. It is ordinary treasury management and it is often the right answer. The objection to this technique is narrower and more specific: it is a way of doing that *without anyone recording that it happened*. The loan does not appear in the fund's borrowings, the drawing does not go through a credit committee, the interest rate is not quoted, the investors are not told, and the position looks identical to an ordinary FX hedge in every system the fund owns. Fix those five things and the technique is defensible. Leave them unfixed and the question is not whether the fund is running a risk it has not measured — it is how large that risk will be before anyone notices.

Reference

A Glossary

All-in rate	The forward rate actually booked on a contract: spot plus forward points. Under a rollover, the all-in rate on the far leg is deliberately off-market.
CLS	Continuous Linked Settlement. The FX settlement system whose payment-versus-payment netting is what allows the two offsetting legs to settle for zero cash.
Covered interest parity	The no-arbitrage relationship fixing the forward rate from spot and the two interest rates. The reason a forward contains no view.
CSA	Credit Support Annex to the ISDA Master. Governs whether, and how much, collateral moves between the parties. Determines whether this technique delivers any benefit at all.
Deferred balance	The cumulative unsettled hedge profit or loss carried inside off-market strikes. The exposure this document is about. Not a standard market term — which is itself part of the problem.
Far leg / near leg	The two dates of an FX swap. In a rollover the near leg reverses the maturing contract at its original rate; the far leg carries the position — and the deferred balance — forward.
Historic (historical) rate rollover	Extending a maturing FX forward by transacting the near leg at the original rather than the prevailing rate, so that no cash settles and the profit or loss is deferred.
PSFX	Physically-settled FX forward. The product class carved out of mandatory variation margin for non-institutional counterparties under EMIR.
Wrong-way risk	Exposure that increases as the counterparty's or the underlying's credit quality deteriorates. Here: the deferred balance is largest in the risk-off conditions that also stress the fund's assets and the dealer's appetite.

B Model, assumptions and sources

Model

All figures are computed from the recursion in section 4, implemented in `hrr_model.py` alongside this document. The v2 additions expose the identities used in Part III directly: `decompose()` (the carry / spot / financing split in section 19), `strike_check()` (the market-data verification in section 20), `breach_grid()` (the calibration in section 21) and `hedge_ratio_view()` (the orthogonality in Figure 11). The limit register `HRR_Limit_Register.xlsx` is generated by `build_register.py` from those same functions, so the two artefacts cannot drift apart; regenerate both after any model change. Base parameters: USD 100m hedged notional; EURUSD spot 1.1000 at inception; USD rate 4.5%; EUR rate 2.5%; dealer accrual spread 75bp over EUR unless stated. Forward rates from covered interest parity throughout. The Monte Carlo uses geometric Brownian motion on EURUSD at 8% annualised volatility with risk-neutral drift equal to the rate differential, 200,000 paths, quarterly rolls over five years.

Known simplifications

- The accrual rate is symmetric in the model. Real dealers charge a spread when you owe and pay below benchmark when you are owed; the true cost is therefore higher than Table 6 across a full cycle. Noted in section 12.
- Constant interest rates. Rate moves change forward points and would add a second-order term to the balance.
- Lognormal FX with constant volatility understates the tail. Realised FX distributions are fat-tailed and trending, which is the specific behaviour that damages this structure — so the section 12 percentiles should be read as a floor, not a central estimate.

- No bid-offer on the roll itself, and no CVA/FVA adjustment beyond the stated spread.
- Scenario paths are illustrative and chosen to sit inside historical experience; they are not forecasts.

Sources

- **FX Global Code**, Global Foreign Exchange Committee — Principle 12 (Execution). The text quoted in section 15 is verbatim from the current Code. globalfx.org/uploads/fx_global.pdf
- **Foreign Exchange Committee (Federal Reserve Bank of New York)**, *Letter on Historical Rate Rollovers: A Dangerous Practice*, and the Committee's 1995 Annual Report treatment. newyorkfed.org/fxc
- **APRA**, *Historical rate rollovers — frequently asked questions*, and Prudential Standard APS 180 paragraph 17. apra.gov.au
- **Ludwig Report** into the Allied Irish Banks / Allfirst foreign exchange losses, 2002.
- **EMIR margin RTS** as amended, on the scope of the variation margin requirement for physically-settled FX forwards and swaps; ESAs draft RTS of 19 December 2017 and the subsequent amending regulation.
- **Source deck**: three slides on Historic Rollover Rate FX Forwards — executive summary, trading process, and the Bloomberg AIM order with worked example. Table 3 reproduces the example as given.

Figures in this document are model output on stated assumptions and are intended to support a risk discussion, not to constitute a valuation, an accounting opinion, a legal opinion or investment advice. The regulatory references should be checked against current text before being cited externally; the FX Global Code in particular is periodically revised and principle numbering has changed between editions.